

Groups, Rings and Modules

1 Revision on IA Groups

Definition 1.1 – Group

Let G be a set and $e \in G$. Let $\cdot : G \times G \rightarrow G$ be a function. A group is a triple $(G, \cdot, e)$ such that

- (i) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$;
- (ii) $a \cdot e = e \cdot a = a$ for all $a \in G$;
- (iii) there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$ for all $a \in G$.

The size of G is called the order of G , written as $|G|$. From IA Groups, we know that inverses are unique.

Definition 1.2 – Subgroup

Let G be a group and $H \subseteq G$ be a subset. We say that H is a subgroup, denoted $H \leq G$, if

- (i) $e \in H$;
- (ii) $a \cdot b \in H$ for all $a, b \in H$;
- (iii) $(H, \cdot, e)$ is a group.

Proposition 1.1 – Subgroup test

A nonempty subset $H \subseteq G$ is a subgroup if and only if $h_1 h_2^{-1} \in H$ for all $h_1, h_2 \in H$.

Definition 1.3 – Abelian group

A group G is abelian if $g_1 g_2 = g_2 g_1$ for all $g_1, g_2 \in G$.

Example 1.1 – Groups

- (i) The number systems $(\mathbb{Z}, +, 0)$, $(\mathbb{Q}, +, 0)$, $(\mathbb{R}, +, 0)$, $(\mathbb{C}, +, 0)$ and $(\mathbb{Z}/n\mathbb{Z}, +, 0)$.
- (ii) The permutations of $\{1, \dots, n\}$ under composition form a group called the symmetric group S_n . More generally, for any set X , there is a group $\text{Sym}(X)$ formed in the same way.
- (iii) The dihedral group D_{2n} is the group of symmetries of a regular n -gon. By labelling the vertices, we can see that $D_{2n} \subseteq S_n$.
- (iv) The general linear group over $\mathbb{R}$, denoted $\text{GL}_n(\mathbb{R}) = \{f : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid f \text{ is a } \mathbb{R}\text{-linear bijection}\} = \{n \times n \text{ invertible matrices over } \mathbb{R}\}$.

Example 1.2 – Subgroups

- (i) The even permutations of S_n form the alternating group $A_n \leq S_n$.
- (ii) The rotational symmetries of an n -gon $C_n \leq D_{2n}$.
- (iii) The matrices with determinant one $SL_n(\mathbb{R}) \leq GL_n(\mathbb{R})$.

Definition 1.4 – Left coset

Let G be a group and $g \in G$. Let $H \leq G$ be a subgroup. The left coset gH is

$$gH = \{x \in G \mid x = gh \text{ for some } h \in H\}.$$

Since $e \in H$, we know that $g \in gH$. Also, the set of all cosets partition G , that is there are no overlaps between different cosets and the union of all cosets is G . Furthermore, the map $h \mapsto gh$ defines a bijection from H to gH so all cosets have the same size.

Theorem 1.1 – Lagrange’s theorem

If G is a finite group and $H \leq G$, then

$$|G| = |H||G : H|$$

where $|G : H|$ is the number of left cosets of H in G .

Definition 1.5 – Order

Let G be a group and $g \in G$. The order of g , written $\text{ord}(g)$, is the smallest positive integer n such that $g^n = e$. By convention, if no such n exists, we take its order to be ∞ .

Proposition 1.2

Let G be a group and $g \in G$. Then, $\text{ord}(g)$ divides $|G|$.

Proof. Let $n = \text{ord}(g)$ and consider $H = \{e, g, g^2, \dots, g^{n-1}\}$. If $n = \infty$, then there is nothing to prove. We can check that $H \leq G$ using the subgroup test. Also, the elements of H are distinct. If $g^i = g^j$ with $i < j$, then $g^{j-i} = e$ with $j - i < n$, contradicting the minimality of n . Applying Lagrange’s theorem gives the result. $\square$

2 Normal subgroups, quotients and homomorphisms

Let $H \leq G$. When are two cosets of H equal? If $gH = g'H$, then $g \in g'H$ and we can write $g = g'h$ for some $h \in H$. Hence, $g(g')^{-1} \in H$ if and only if $gH = g'H$.

We denote the set of left cosets of $H \leq G$ by $G/H = \{gH \mid g \in G\}$. Does G/H carry a natural group structure? We propose the formula $(g_1H)(g_2H) = (g_1g_2)H$. We need to check that this rule does not depend on the way we write the cosets g_1H and g_2H . If $g_2H = g'_2H$, then $g'_2 = g_2h$ for some $h \in H$. We verify that $(g_1H)(g'_2H) = (g_1g'_2)H = g_1g_2hH = g_1g_2H$. If $g_1H = g'_1H$, then $g'_1 = g_1h$ for some $h \in H$. We need $g_1g_2H = g_1hg_2H$. This is not automatic, but is true if $g_2^{-1}hg_2 \in H$. We need this to hold for all $g_2 \in G$ and $h \in H$ in order for the proposed formula to define a group structure.

Definition 2.1 – Normal subgroup

A subgroup $H \leq G$ is normal if $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$. We denote this by $H \triangleleft G$.

When $H \triangleleft G$, then G/H inherits a group structure by the rule above. We refer to this as the quotient group of G by H . The identity element is H .

Definition 2.2 – Homomorphism

Let G and H be groups. A homomorphism from G to H is a function $f : G \rightarrow H$ such that

$$f(g_1g_2) = f(g_1)f(g_2)$$

for all $g_1, g_2 \in G$.

It follows that $f(e_G) = e_H$.

Lemma 2.1

If $f : G \rightarrow H$ is a homomorphism, then $f(g^{-1}) = f(g)^{-1}$.

Proof. $f(g)f(g^{-1}) = f(gg^{-1}) = f(e_G) = e_H$. Hence, $f(g^{-1}) = f(g)^{-1}$ by the uniqueness of inverses. $\square$

Definition 2.3 – Kernel and image

Let $f : G \rightarrow H$ be a homomorphism. The kernel of f is

$$\ker(f) = \{g \in G \mid f(g) = e_H\}.$$

The image of f is

$$\operatorname{im}(f) = \{h \in H \mid h = f(g) \text{ for some } g \in G\}.$$

Proposition 2.1

Let $f : G \rightarrow H$ be a homomorphism. Then, $\ker(f) \triangleleft G$ and $\operatorname{im}(f) \leq H$.

Proof. Note that $e \in \ker(f)$ so $\ker(f)$ is nonempty. If $x, y \in \ker(f)$, then $f(xy^{-1}) = f(x)f(y)^{-1} = e$ so $xy^{-1} \in \ker(f)$. By the subgroup test, $\ker(f) \leq G$. For normality, let $g \in G$ and $x \in \ker(f)$. $f(gxg^{-1}) = f(g)f(x)f(g)^{-1} = f(g)f(g)^{-1} = e$ so $gxg^{-1} \in \ker(f)$ and $\ker(f) \triangleleft G$.

Let $a, b \in \operatorname{im}(f)$ and say $a = f(x)$ and $b = f(y)$. Then, $ab^{-1} = f(x)f(y)^{-1} = f(xy^{-1})$ so $ab^{-1} \in \operatorname{im}(f)$. By the subgroup test, $\operatorname{im}(f) \leq H$. $\square$

Observe that $G \rightarrow G/H$ sending $g \mapsto gH$ is a homomorphism when $H \triangleleft G$.

Definition 2.4 – Isomorphism

A homomorphism $f : G \rightarrow H$ is an isomorphism if it is a bijection. Two groups are said to be isomorphic if there exists an isomorphism between them, written as $G \cong H$.

Theorem 2.1 – First isomorphism theorem

Let $f : G \rightarrow H$ be a homomorphism. Then, $\ker(f) \triangleleft G$ and the homomorphism

$$\begin{aligned} \varphi : G/\ker(f) &\rightarrow \operatorname{im}(f) \\ g\ker(f) &\mapsto f(g) \end{aligned}$$

is an isomorphism. In particular, $G/\ker(f) \cong \operatorname{im}(f)$.

Proof. $\ker(f) \triangleleft G$ was shown in Proposition 1.3. First, we check that φ is well-defined. If $g \ker(f) = g' \ker(f)$, then $g'g^{-1} \in \ker(f)$. So, $f(g'g^{-1}) = e$ which gives $f(g) = f(g')$. For surjectivity, let $h \in \text{im}(f)$, then $h = f(g)$ for some $g \in G$ and $h = \varphi(g \ker(f))$. For injectivity, if $\varphi(g \ker(f)) = \varphi(g' \ker(f))$, then $f(g) = f(g')$ so $g'g^{-1} \in \ker(f)$ and $g \ker(f) = g' \ker(f)$. $\square$

Example 2.1

Consider the homomorphism

$$\begin{aligned} \exp : \mathbb{C}^+ &\rightarrow \mathbb{C}^\times \\ z &\mapsto e^z. \end{aligned}$$

The kernel of $\exp$ is $2\pi i\mathbb{Z}$. The first isomorphism theorem gives that $\mathbb{C}^\times \cong \mathbb{C}^+ / (2\pi i\mathbb{Z})$.

Theorem 2.2 – Second isomorphism theorem

Let $H \leq G$ and $K \triangleleft G$. Then,

$$HK = \{hk \mid h \in H \text{ and } k \in K\} \leq G$$

and $H \cap K \triangleleft H$ and there is an isomorphism

$$(HK)/K \cong H/(H \cap K).$$

Proof. First, we prove that $HK \leq G$. It contains e so it is nonempty. Consider $x = hk$ and $y = h'k'$. We will show that $yx^{-1} \in HK$.

$$yx^{-1} = h'k'(hk)^{-1} = h'k'k^{-1}h^{-1} = (h'h^{-1})h(k'k^{-1})h^{-1} = h'h^{-1}k'' \in HK$$

for some $k'' \in K$ by the normality of K . We then apply the subgroup test.

$H \cap K$ is a kernel of the homomorphism $\varphi : H \rightarrow G/K$ (restricting the domain of $G \rightarrow G/K$) so it is a normal subgroup of H . Also, we can apply the first isomorphism theorem to φ . This gives $\text{im}(\varphi) \cong H/(H \cap K)$. Notice that $\text{im}(\varphi)$ is the set of K -cosets in G that are representable by elements of H , which is $\{hK \mid h \in H\} = (HK)/K$. $\square$

Proposition 2.2 – Correspondence between subgroups of G/K and of G

Let $K \triangleleft G$. There is a bijection

$$\begin{aligned} \{\text{subgroups of } G \text{ that contains } K\} &\leftrightarrow \{\text{subgroups of } G/K\} \\ K \triangleleft L \leq G &\mapsto L/K \leq G/K \\ \{g \in G \mid gK \in A\} &\leftrightarrow A \leq G/K. \end{aligned}$$

The fact that these are inverses is a tedious but straightforward check. Also, the correspondence matches normal subgroups on the two sides.

Theorem 2.3 – Third isomorphism theorem

Let $K \leq L \leq G$ be normal subgroups of G . Then, there exists an isomorphism $(G/K)/(L/K) \cong G/L$.

Proof. Define a map

$$\begin{aligned} \varphi : G/K &\rightarrow G/L \\ gK &\mapsto gL. \end{aligned}$$

φ is well-defined. If $gK = g'K$, then $g'g^{-1} \in K \subseteq L$ so $gL = g'L$. It is clearly a homomorphism. We can calculate $\ker(\varphi) = \{gK \mid gL = L\} = \{gK \mid g \in L\} = L/K$. By the first isomorphism theorem, we are done. $\square$

Definition 2.5 – Simple group

A group G is called simple if the only normal subgroups of G are $\{e\}$ and G .

Proposition 2.3

Let G be an abelian group. G is simple if and only if $G \cong C_p$ for some prime p .

Proof. If $G \cong C_p$, then any non-identity element generates G so G is simple. If G is simple and abelian, take $e \neq g \in G$ and consider $\{g^n \mid n \in \mathbb{Z}\}$. This is a subgroup. G is abelian so normality is automatic, and by simplicity G is cyclic. Suppose G is infinite and cyclic, then $G \cong \mathbb{Z}$ but not simple. Now suppose that G is finite and $G \cong C_m$. If q divides m , take the subgroup generated by $g^{m/q}$ where g generates G . Normality is automatic so $q = 1$ or m by simplicity. Hence, m is prime. $\square$

Theorem 2.4

Let G be a finite group. There exist subgroups

$$G = H_1 \triangleright H_2 \triangleright H_3 \triangleright \cdots \triangleright H_n = \{e\}$$

such that H_i/H_{i+1} is simple.

Proof. If G is simple, take $H_2 = \{e\}$ and we are done. Otherwise, let H_2 be a proper normal subgroup of maximal order in G . We claim that G/H_2 is simple. If not, consider $\pi : G \rightarrow G/H_2$. If $N \triangleleft G/H_2$ is a proper normal subgroup, then we can find a proper normal subgroup in G that contains H_2 by Proposition 2.2. But this contradicts the maximality of $|H_2|$. Hence, G/H_2 is simple. We can repeat the argument by replacing G with H_2 . Note that $H_{i+1} \neq H_i$ so $|H_{i+1}| \leq |H_i|$ and the process terminates. $\square$

3 Group actions and permutations

Let X be a set and $\text{Sym}(X) = \{f : X \rightarrow X \mid f \text{ is a bijection}\}$ be the set of permutations of X . The identity element is the identity function and the group operation is composition. We write S_n for $\text{Sym}(\{1, \dots, n\})$.

From IA Groups, we can write $\sigma \in S_n$ as products of disjoint cycles. Also, every $\sigma \in S_n$ is a product of transpositions. The parity of the number of such transpositions is the sign of σ .

$$\text{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is a product of an even number of transpositions} \\ -1 & \text{if } \sigma \text{ is a product of an odd number of transpositions.} \end{cases}$$

Note that sgn is a homomorphism, and is surjective if $n \geq 3$.

Let X be a set and G be a group. Consider a homomorphism $G \rightarrow \text{Sym}(X)$. This is called a permutation representation of G .

Definition 3.1 – Group action

An action of a group G on a set X is a function $G \times X \rightarrow X$ such that

$$(g, x) \mapsto g * x$$

- (i) $e * x = x$ for all $x \in X$;
- (ii) $g_1 * (g_2 * x) = (g_1 g_2) * x$ for all $g_1, g_2 \in G$ and $x \in X$.

This is denoted by $G \curvearrowright X$.

Proposition 3.1

The map

$$\begin{aligned} a : \{\text{homomorphisms } G \rightarrow \text{Sym}(X)\} &\rightarrow \{\text{actions } G \curvearrowright X\} \\ \varphi : G \rightarrow \text{Sym}(X) &\mapsto a(\varphi) : G \times X \rightarrow X \\ &\quad (g, x) \mapsto \varphi(g) \cdot x \end{aligned}$$

is a bijection.

Proof. First note that $a(\varphi)$ is a group action by the homomorphism property. We construct an inverse to a . Let $*$: $G \times X \rightarrow X$ be an action. Define

$$\begin{aligned} \varphi : G &\rightarrow \text{Sym}(X) \\ g &\mapsto \varphi(g) = [g * () : X \rightarrow X]. \end{aligned}$$

This is a permutation as $g^{-1} * () : X \rightarrow X$ is the inverse. Indeed,

$$\varphi(g^{-1})(\varphi(g)(x)) = g^{-1} * (g * x) = (g g^{-1}) * x = e * x = x$$

for all $x \in X$. Similarly, $\varphi(g) \circ \varphi(g^{-1}) = \text{id}_X$. To see this rule gives a homomorphism,

$$\varphi(g_1)(\varphi(g_2)(x)) = g_1 * (g_2 * x) = (g_1 g_2) * x = \varphi(g_1 g_2)(x)$$

for all $g_1, g_2 \in G$ and $x \in X$. Hence, $\varphi(g_1 g_2) = \varphi(g_1) \circ \varphi(g_2)$. This construction inverts a . $\square$

For an action $G \curvearrowright X$ given by $\varphi : G \rightarrow \text{Sym}(X)$, we denote $G^X = \text{im}(\varphi) \leq \text{Sym}(X)$ and $G_X = \ker(\varphi) \triangleleft G$. By the first isomorphism theorem, $G^X \cong G/G_X$.

Example 3.1

Let G be the symmetry group of the unit cube. Let X be the set of diagonals. G acts on X . Consider the associated $\varphi : G \rightarrow \text{Sym}(X) \cong S_4$. We can check that $\text{im}(\varphi) \cong S_4$. Also, $\ker(\varphi) = \{\text{id}, \text{sending every vertex to its opposite}\} \cong C_2$. Hence, $|G| = |S_4| |C_2| = 48$ by the first isomorphism theorem.

Example 3.2

For any group G , we can make G act on the set underlying G by

$$\begin{aligned} G \times G &\rightarrow G \\ (g, g_1) &\mapsto g g_1. \end{aligned}$$

This gives $\varphi : G \rightarrow \text{Sym}(G)$. Note that $\ker(\varphi) = \{e\}$ so φ is injective. Hence, G is isomorphic to a subgroup of $\text{Sym}(G)$. The takeaway is that symmetric groups are very complicated.

Example 3.3

Let $H \leq G$ and $X = G/H$ be the set of left cosets. Let $G \curvearrowright X$ by $(g, g_1H) \mapsto gg_1H$. We can check that this indeed defines an action. This gives $\varphi : G \rightarrow \text{Sym}(X)$. It is obvious that $\ker(\varphi) \subseteq H$ but more is true. If $g \in \ker(\varphi)$, then $gg_1H = g_1H$ for all $g_1 \in G$ so $g_1^{-1}gg_1 \in H$ and $\ker(\varphi) \subseteq \bigcap_{g_1 \in G} g_1Hg_1^{-1}$.

Note that this containment is reversible. If $g \in \bigcap_{g_1 \in G} g_1Hg_1^{-1}$, then for each $g_1 \in G$, we have $g_1gg_1^{-1} \in H$ so $gg_1H = g_1H$ and $g \in \ker(\varphi)$. Hence, $\ker(\varphi) = \bigcap_{g_1 \in G} g_1Hg_1^{-1}$.

Theorem 3.1

Let G be finite and $H \leq G$ of index n . There exists $K \triangleleft G$ with $K \leq H$ such that G/K is isomorphic to a subgroup of S_n . Hence, $|G/K|$ divides $n!$ and $|G/K| \geq n$.

Proof. Consider $\varphi : G \rightarrow \text{Sym}(G/H)$ from Example 3.3. The kernel $K = \ker(\varphi)$ is normal. We have shown that it is contained in H already. The first isomorphism theorem gives $G/K \cong \text{im}(\varphi) \leq \text{Sym}(G/H) \cong S_n$. Lagrange gives $|G/K|$ divides $n!$. Also, $K \leq H$ so $|G/K| \geq |G/H| = n$. $\square$

Corollary 3.1

Let G be a non-abelian simple group. Let $H \leq G$ be a proper subgroup with index $n > 1$. Then, $n \geq 5$ and G is isomorphic to a subgroup of A_n .

Proof. Consider the action of G on the set G/H by left multiplication. We get $\varphi : G \rightarrow \text{Sym}(G/H) \cong S_n$. By simplicity, $\ker(\varphi) = \{e\}$ or G , but $\ker(\varphi) \neq G$ because H is proper. So, φ is injective. By the first isomorphism theorem, $G \cong \text{im}(\varphi) \leq \text{Sym}(G/H) \cong S_n$. So, we know $G \leq S_n$ and now we want $G \cap A_n = G$. We know that $A_n \triangleleft S_n$ so $A_n \cap G \triangleleft G$ by the second isomorphism theorem. Since G is simple, the only possibilities are $G \cap A_n = \{e\}$ or G . We need to rule out the first option. If $G \cap A_n = \{e\}$, by the second isomorphism theorem, $G \cong G/(G \cap A_n) \cong (GA_n)/A_n \leq S_n/A_n$ which is of order two. Since G is non-abelian, this leads to a contradiction. $\square$

Definition 3.2 – Orbit and stabiliser

Let $G \curvearrowright X$ be a group action. The orbit of $x \in X$ is

$$Gx = \{gx \mid g \in G\} \subseteq X.$$

The stabiliser of $x \in X$ is

$$G_x = \{g \in G \mid gx = x\} \leq G.$$

Theorem 3.2

For $G \curvearrowright X$ and $x \in X$, there is a bijection $Gx \rightarrow G/G_x$. If G is finite, then we have $|G| = |Gx||G_x|$.

4 Conjugacy, centralisers and normalisers

A group G can act on itself by conjugation as follows.

$$\begin{aligned} G \times G &\rightarrow G \\ (g, h) &\mapsto ghg^{-1} \end{aligned}$$

This is equivalent to a homomorphism $G \rightarrow \text{Sym}(G)$. If we fix $g \in G$, the permutation $G \rightarrow G$ given by $h \mapsto ghg^{-1}$ is a homomorphism from G to itself.

Definition 4.1 – Automorphism

Let G be a group. A permutation $a : G \rightarrow G$ is an automorphism if it is a homomorphism. The set of all automorphisms of G

$$\text{Aut}(G) = \{f : G \rightarrow G \mid f \text{ is an automorphism}\} \leq \text{Sym}(G)$$

is a subgroup.

Example 4.1

Let $G = \mathbb{R}^2$ with coordinate-wise addition. $\text{Aut}(G)$ contains $\text{GL}_2(\mathbb{R})$ as a subgroup.

Definition 4.2 – Conjugacy class, centraliser and centre

Fix $g \in G$. The conjugacy class of g is the orbit under conjugation action. That is,

$$\text{ccl}_G(g) = \{hgh^{-1} \mid h \in G\}.$$

The centraliser of g is the stabiliser under conjugation action. That is,

$$C_G(g) = \{h \in G \mid hgh^{-1} = g\}.$$

The centre of G is the kernel of the conjugation action. That is,

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}.$$

Proposition 4.1

Let G be a finite group. Then, $|\text{ccl}_G(x)| = |G : C_G(x)| = |G|/|C_G(x)|$ for all $x \in G$.

Proof. Apply orbit-stabiliser to the conjugation action. □

Definition 4.3 – Normaliser

Let $H \leq G$. The normaliser of H in G is $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$.

Observe that $H \triangleleft N_G(H)$. In fact by definition, $N_G(H)$ is the largest subgroup of G which H is normal in.

5 Simplicity of the alternating groups A_n with $n \geq 5$

Recall from IA Groups that a conjugacy class in S_n is the set of all elements of a fixed cycle type.

Theorem 5.1

A_n with $n \geq 5$ is simple.

Proof. The theorem can be proved by the following three steps.

- (i) A_n is generated by 3-cycles.
- (ii) Any $H \triangleleft A_n$ that contains a 3-cycle contains all of them.
- (iii) Any non-trivial $H \triangleleft A_n$ contains a 3-cycle.

For (i), note that any $g \in A_n$ is a product of an even number of transpositions. Consider pairs of transpositions. There are three possibilities. $(a\ b)(a\ b) = e$ or $(a\ b)(b\ c) = (a\ b\ c)$ or $(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d)$. All of these are products of 3-cycles and we are done.

For (ii), note that any two 3-cycles are conjugate when viewed in S_n . Let δ and δ' be 3-cycles so we can write $\delta' = \sigma\delta\sigma^{-1}$ for some $\sigma \in S_n$. Since $n \geq 5$, there exists a transposition τ disjoint from δ so commutes with δ . Then, we can write $\delta' = (\sigma\tau)\delta(\sigma\tau)^{-1}$ if $\sigma \in S_n \setminus A_n$. By the normality of H , we are done.

For (iii), there are several cases to consider.

- (a) If H contains an element of the form $\sigma = (1\ 2\ \dots\ r)\tau$ where τ is disjoint from $\{1, \dots, r\}$ and $r \geq 4$. By the normality of H , if $\delta = (1\ 2\ 3)$, then $\delta^{-1}\sigma\delta \in H$ so $\sigma^{-1}\delta^{-1}\sigma\delta \in H$. Notice that τ commutes with δ and also commutes with $(1\ 2\ \dots\ r)$. We expand $\sigma^{-1}\delta^{-1}\sigma\delta = (r\ \dots\ 2\ 1)(1\ 3\ 2)(1\ 2\ \dots\ r)(1\ 2\ 3) = (2\ 3\ r)$ and we have found a 3-cycle.
- (b) Suppose H contains an element of the form $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$ in the disjoint cycle decomposition. Take $\delta = (1\ 2\ 4)$ and calculate $\sigma^{-1}\delta^{-1}\sigma\delta$. This is a 5-cycle so we can apply (a).
- (c) Suppose H contains $\sigma = (1\ 2\ 3)\tau$ where τ is a product of disjoint transpositions. If τ contains anything larger, we can relabel and apply (a) or (b). Note that $\sigma^2 = (1\ 2\ 3)^2$ which is a 3-cycle.
- (d) Suppose H contains $\sigma = (1\ 2)(3\ 4)\tau$ where τ is a product of transpositions. Let $\delta = (1\ 2\ 3)$ and consider $\mu = \sigma^{-1}\delta^{-1}\sigma\delta = (1\ 4)(2\ 3)$. Let $\nu = (1\ 5\ 2)\mu(1\ 2\ 5) = (1\ 3)(4\ 5)$. Since $\mu\nu = (1\ 2\ 3\ 4\ 5) \in H$, then H contains a 3-cycle by (a).

Up to relabelling $\{1, \dots, n\}$ by a permutation, this covers all the cases. □

6 Finite p -groups

Let p be a prime. Groups of order p^n for some positive integer n are often called p -groups.

Theorem 6.1

Let G be a p -group. The centre of G is non-trivial.

Proof. Consider G acting on itself by conjugation. The centre $Z(G)$ is the set of all size one orbits. The orbits partition G so

$$|G| = p^n = |Z(G)| + \sum (\text{sizes of conjugacy classes of size } > 1).$$

We know that conjugacy classes have sizes that divide p^n . If its size is bigger than one, then p divides its size. Therefore, p divides the summation above. Since $e \in Z(G)$, we know $|Z(G)| \geq 1$. Hence, $|Z(G)| \geq p$. □

Lemma 6.1

If $G/Z(G)$ is cyclic, then G is abelian.

Proof. There is a coset $xZ(G) \in G/Z(G)$ that generates so every coset has the form $x^mZ(G)$. This means that every $g \in G$ can be written as $g = x^mz$ for some $z \in Z(G)$. If $g = x^mz$ and $h = x^nz'$, then $gh = x^mzx^nz' =$

$$x^{m+n}zz' = x^n z' x^m z = hg.$$

□

Theorem 6.2

A group of size p^2 is always abelian.

Proof. By Theorem 6.1, $|Z(G)|$ is either p or p^2 . If $|Z(G)| = p^2$, then we are done. If $|Z(G)| = p$, then $|G/Z(G)| = p$ so $G/Z(G) \cong C_p$. By Lemma 6.1, we are done. □

Theorem 6.3

Let G be a p -group of size p^n . Then, G has a subgroup of size p^k for all $k \in \{0, 1, \dots, n\}$.

Proof. We proceed by induction on n . The base case of $n = 1$ is trivial. Now let $n > 1$. We can take $\{e\}$ for $k = 0$. Otherwise, note that $Z(G)$ is non-trivial so we can let $x \in Z(G)$ with $x \neq e$. $\text{ord}(x)$ is a power of p . If we replace x with a power of itself, we can assume it has order p . So, $\langle x \rangle$ has size p and is normal since $x \in Z(G)$. Consider $G/\langle x \rangle$. This is a group of size p^{n-1} . By induction, it contains a subgroup $L \leq G/\langle x \rangle$ of size p^{k-1} . By the correspondence of subgroups, we can find some $K \leq G$ containing $\langle x \rangle$ such that $K/\langle x \rangle \cong L$. But then $|K| = p^k$. □

7 Finite abelian groups

Theorem 7.1 – Classification of finite abelian groups

Let G be a finite abelian group. There exist positive integers $d_1, \dots, d_r$ such that

$$G \cong C_{d_1} \times \cdots \times C_{d_r}.$$

Moreover, we can choose d_i such that $d_{i+1} \mid d_i$, in which case the expression is unique.

Example 7.1

The abelian groups of order 8 are exactly C_8 , $C_4 \times C_2$ and $C_2 \times C_2 \times C_2$.

Lemma 7.1

If n and m are coprime, then $C_n \times C_m \cong C_{nm}$.

Proof. It suffices to find an element of order nm in $C_n \times C_m$. If $g \in C_n$ is of order n and $h \in C_m$ is of order m , consider $(g, h) \in C_n \times C_m$. Suppose it has order k , then $(g, h)^k = (g^k, h^k) = (e, e)$. So, $n \mid k$ and $m \mid k$. Since n and m are coprime, $nm \mid k$. By Lagrange, $k \mid nm$ hence $k = nm$. □

Corollary 7.1 – Primary decomposition

Let G be a finite abelian group. $G \cong C_{l_1} \times \cdots \times C_{l_k}$ where each l_i are prime powers.

Proof. Combine Theorem 7.1 with Lemma 7.1. □

8 Sylow theorems

Definition 8.1 – Sylow p -subgroup

Let G be a finite group of size $p^a m$ where $p \nmid m$ and p is some prime. A Sylow p -subgroup of G is a subgroup of order p^a .

Theorem 8.1 – Sylow theorems

Let G be a finite group and p be a prime.

- (i) The set $\text{Syl}_p(G) = \{P \leq G \mid P \text{ is a Sylow } p\text{-subgroup}\}$ is nonempty.
- (ii) Any $H, H' \in \text{Syl}_p(G)$ are conjugate. That is, $H' = gHg^{-1}$ for some $g \in G$.
- (iii) If $n_p = |\text{Syl}_p(G)|$, then $n_p \equiv 1 \pmod{p}$ and n_p divides $|G|$ hence m .

Proof. For (i), we try to produce a Sylow p -subgroup as a stabiliser of an element of an appropriate action. Let $|G| = n = p^a m$ where $p \nmid m$. Define $\Omega = \{X \subseteq G \mid |X| = p^a\}$ be the set of all subsets of G with size p^a . Let G act on Ω by $g * \{g_1, \dots, g_{p^a}\} = \{gg_1, \dots, gg_{p^a}\}$.

Claim 8.1 – We have $|\Omega| = \binom{n}{p^a} = m \not\equiv 0 \pmod{p}$.

Proof. As an exercise. □

Claim 8.2 – Suppose $U \in \Omega$ and $H \leq G$ stabilises U , then $|H| \mid |U|$.

Proof. H stabilises U so $hU = U$ for all $h \in H$. In other words, for each $u \in U$, the coset Hu is contained in U . Every $u \in U$ lies in some coset of H and the cosets are disjoint. Since the cosets partition U , we get $|H| \mid |U|$. □

Let O_i be the orbits of the action. $|G| = |O_1| + \dots + |O_r|$ since the orbits partition. Since $p \nmid |G|$, there exists an orbit Θ whose size is coprime to p . Let $T \in \Theta$. By orbit-stabiliser, $p^a m = |G| = |\Theta| |\text{Stab}(T)|$. By Claim 8.2, $|\text{Stab}(T)|$ divides p^a . We are done because no factors of p are in $|\Theta|$ so $|\text{Stab}(T)| = p^a$.

For (ii), we will show the following stronger statement.

Claim 8.3 – If $Q \leq G$ is a p -group in G and $P \in \text{Syl}_p(G)$, then there exists $g \in G$ such that $gQg^{-1} \subseteq P$.

Proof. Let Q act on G/P by $q * gP = qgP$. The size of an orbit of this action is either one or is divisible by p . Since $|G/P|$ is not divisible by p , there is at least one size one orbit. In other words, there exists some $g \in G$ such that $qgP = gP$ for all $q \in Q$. Rearranging, $g^{-1}qg \in P$ for all $q \in Q$. Hence, $gQg^{-1} \subseteq P$. □

For (iii), consider the conjugation action of G on $\text{Syl}_p(G)$. It has only one orbit. The size of that orbit is n_p so $n_p \mid |G|$. For $n_p \equiv 1 \pmod{p}$, let $P \in \text{Syl}_p(G)$. Consider P acting on $\text{Syl}_p(G)$ by conjugation. The size of an orbit of this action is either one or is divisible by p . $\{P\}$ is a size one orbit so it suffices to show that all other orbits have size divisible by p . Say $\{Q\}$ is another size one orbit. For every $h \in P$, we would have $hQh^{-1} = Q$. Thus, P is contained in the normaliser $N_G(Q)$. Observe that $Q \leq N_G(Q) \leq G$ so $Q \in \text{Syl}_p(N_G(Q))$. But Q is normal in $N_G(Q)$ and P is a Sylow p -subgroup of $N_G(Q)$, hence $Q = P$. □

Lemma 8.1

If $\text{Syl}_p(G) = \{P\}$ has size one, then P is normal.

Proof. For any $g \in G$, gPg^{-1} is isomorphic as a group to P so it is a Sylow p -subgroup. Hence, $gPg^{-1} = P$ and P is normal. $\square$

Corollary 8.1

Let G be a non-abelian simple group and $p \mid |G|$. Then, $|G|$ divides $n_p!/2$ and $n_p \geq 5$.

Proof. The conjugation action of G on $\text{Syl}_p(G)$ gives a homomorphism $\varphi : G \rightarrow \text{Sym}(\text{Syl}_p(G)) \cong S_{n_p}$. By the simplicity of G , $\ker(\varphi) = \{e\}$ or $\ker(\varphi) = G$. If $\ker(\varphi) = G$, then $gPg^{-1} = P$ for all $g \in G$. That is, P is normal and $P = G$ since $P \neq \{e\}$. But this is impossible because $Z(P) \triangleleft P$ is non-trivial and so contradicts that G is simple. So, $\ker(\varphi) = \{e\}$ and φ is injective. Hence, G is isomorphic to a subgroup of S_{n_p} .

Consider $\text{sgn} \circ \varphi : G \rightarrow \{\pm 1\}$. If the composition is surjective, then $\ker(\text{sgn} \circ \varphi)$ is an index two subgroup of G . But G is simple, so this is impossible. Hence, $\text{im}(\varphi) \subseteq A_{n_p}$ and we can conclude by Lagrange. Finally, all non-abelian subgroups of $A_1, \dots, A_4$ are not simple. $\square$

Example 8.1

Let G have size $11 \times 12 = 132$. If G is simple, then none of the n_p 's are equal to one by Lemma 8.1. Since $n_{11} = 1 \pmod{11}$ and $n_{11} \mid 12$, we must have exactly 12 Sylow 11-subgroups. Similarly, $n_3 = 1 \pmod{3}$ and $n_3 \mid 44$, we have $n_3 = 4$ or 22. By Corollary 8.1, $n_3 = 22$. Any two Sylow 11-subgroups intersect only at $\{e\}$ so we already have 120 such elements. There are also at least 44 elements of order 3. Since $|G| = 132$, there is not enough room so G cannot be simple.

Example 8.2

Let $G = \text{GL}_n(\mathbb{Z}/p\mathbb{Z}) = \{n \times n \text{ matrices over } \mathbb{Z}/p\mathbb{Z} \text{ that are invertible}\}$. What is $|G|$? Consider building an invertible matrix column-wise $(\mathbf{v}_1 \mid \mathbf{v}_2 \mid \dots \mid \mathbf{v}_n)$. There are $(p^n - 1)$ choices for $\mathbf{v}_1$ since $\mathbf{v}_1 \neq 0$. There are $(p^n - p)$ choices for $\mathbf{v}_2$ since $\mathbf{v}_2 \notin \langle \mathbf{v}_1 \rangle$. There are $(p^n - p^2)$ choices for $\mathbf{v}_3$ since $\mathbf{v}_3 \notin \langle \mathbf{v}_1, \mathbf{v}_2 \rangle$, and so on. Hence, $|G| = (p^n - 1)(p^n - p) \dots (p^n - p^{n-1})$. The largest power of p dividing $|G|$ is $p^{n(n-1)/2}$. For example, the set of all upper-triangular matrices with ones on the diagonal is a Sylow p -subgroup.

9 Rings

Definition 9.1 – Ring

A ring is a quintuple $(R, +, \cdot, 0_R, 1_R)$ where R is a set with $0_R, 1_R \in R$ and $+, \cdot : R \times R \rightarrow R$ are addition and multiplication respectively satisfying

- (i) $(R, +, 0_R)$ is an abelian group;
- (ii) the operation $\cdot$ is associative, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ and $1_R \cdot a = a \cdot 1_R = a$ for all $a \in R$;
- (iii) multiplication distributes over addition, $r_1 \cdot (r_2 + r_3) = r_1 \cdot r_2 + r_1 \cdot r_3$ and $(r_1 + r_2) \cdot r_3 = r_1 \cdot r_3 + r_2 \cdot r_3$ for all $r_1, r_2, r_3 \in R$.

We will simply say R is a ring instead of referring to the quintuple. The symbol $-r$ will mean the additive

inverse of r . We will write $r - s$ for $r + (-s)$. In this course, we will assume that R is commutative, so $r_1 r_2 = r_2 r_1$ for all $r_1, r_2 \in R$.

Definition 9.2 – Subring

A subring of a ring R is a subset $S \subseteq R$ such that

- (i) $0_R \in S$ and $1_R \in S$;
- (ii) if $s, s' \in S$, then $s + s' \in S$ and $ss' \in S$;
- (iii) $(S, +, \cdot, 0_R, 1_R)$ is a ring.

We denote this by $S \leq R$.

Example 9.1

- (i) The usual number systems $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ are all rings.
- (ii) The Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \leq \mathbb{C}$ is a ring. Similarly, $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\} \leq \mathbb{R}$ is a ring.
- (iii) The dual numbers $\mathbb{C}\langle\varepsilon\rangle = \{a + b\varepsilon \mid a, b \in \mathbb{C}\}$ with operations $(a + b\varepsilon) + (c + d\varepsilon) = (a + c) + (b + d)\varepsilon$ and $(a + b\varepsilon) \cdot (c + d\varepsilon) = ac + (ad + bc)\varepsilon$ is a ring. In particular, $\varepsilon^2 = 0$ with $\varepsilon \neq 0$.

Definition 9.3 – Unit

An element $u \in R$ is a unit if there exists $v \in R$ such that $uv = 1$.

Note that this notion does not interact well with subrings. For example, consider $2 \in \mathbb{Z} \subseteq \mathbb{Q}$. 2 is a unit in $\mathbb{Q}$ but not in $\mathbb{Z}$.

Definition 9.4 – Field

A ring R is a field if every nonzero element is a unit.

Let R be a ring. Notice that $0_R + 0_R = 0_R$. For any $r \in R$, we can multiply and distribute, so $r0_R + r0_R = r0_R$ which simplifies to $r0_R = 0_R$. In particular, if $1_R = 0_R$, then for any $r \in R$, we have $r = 0_R$. Hence, $R = \{0\}$ and is called the zero ring.

Definition 9.5 – Direct product

Let R and S be rings. The direct product $R \times S$ is a ring by

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2) \quad (r_1, s_1) \cdot (r_2, s_2) = (r_1 r_2, s_1 s_2)$$

with $0_{R \times S} = (0_R, 0_S)$ and $1_{R \times S} = (1_R, 1_S)$.

We can check that this indeed defines a ring.

Definition 9.6 – Polynomial

Let R be a ring. A polynomial in X with coefficients in R is an expression

$$f(X) = a_0 + a_1X + \cdots + a_nX^n$$

where $a_i \in R$ and X^i are formal symbols. As convention, we identify $f(X)$ with $f(X) + 0 \cdot X^{n+1}$.

The largest n such that $a_n \neq 0$ is called the degree of $f(X)$. We set the degree of 0 to be $-\infty$. A polynomial is called monic if it has degree n and $a_n = 1$.

Definition 9.7 – Polynomial ring

The polynomial ring $R[X]$ is

$$R[X] = \{f(X) \mid f(X) \text{ is a polynomial with coefficients in } R\}$$

with operations defined the obvious way.

For $f(X) \in R[X]$, we get a function $R \rightarrow R$ defined by $r \mapsto f(r)$.

Definition 9.8 – Ring of formal power series

The ring of formal power series over R in the variable X is

$$R[[X]] = \left\{ \sum_{i=0}^{\infty} r_i X^i \mid r_i \in R \right\}$$

with the standard operations.

Example 9.2

Consider the element $(1 - x)$ in $R[X]$. It is a not unit. If $g(x)(1 - x) = 1$ and $g(x) = a_0 + a_1x + \cdots + a_nx^n$, then $(1 - x)g(x) = a_0 + (a_1 - a_0)x + \cdots + (a_n - a_{n-1})x^n - a_nx^{n+1}$. But this is clearly not 1 by comparing the x^{n+1} coefficient. However, if $(1 - x) \in R[[X]]$, then we have an inverse $(1 - x)(1 + x + x^2 + \cdots) = 1$.

Definition 9.9 – Ring of Laurent polynomials

We write $R[X, X^{-1}]$ for the ring of Laurent polynomials. That is,

$$f = \sum_{i \in \mathbb{Z}} a_i X^i$$

with $a_i \in R$ and $a_i = 0$ for all but finitely many i .

10 Homomorphisms, ideals and quotients

Definition 10.1 – Ring homomorphism

Let R and S be rings. A function $f : R \rightarrow S$ is a ring homomorphism if for all $r_1, r_2 \in R$, we have

- (i) $f(r_1 + r_2) = f(r_1) + f(r_2)$;
- (ii) $f(0) = 0$;
- (iii) $f(r_1 r_2) = f(r_1) f(r_2)$;
- (iv) $f(1) = 1$.

Note that (ii) is implied by (i), but (iv) is not implied by (iii).

Definition 10.2 – Ring isomorphism, kernel and image

If $f : R \rightarrow S$ is a bijective ring homomorphism, it is called an isomorphism. Note that its inverse is also an isomorphism.

The kernel of f is $\ker(f) = \{r \in R \mid f(r) = 0\}$. The image is $\operatorname{im}(f) = \{s \in S \mid f(r) = s \text{ for some } r \in R\}$.

Lemma 10.1

A homomorphism $f : R \rightarrow S$ is injective if and only if $\ker(f) = \{0\}$.

Proof. Implied by the corresponding statement about group homomorphisms. □

Definition 10.3 – Ideal

A subset $I \subseteq R$ is called an ideal, written $I \triangleleft R$, if it satisfies the following.

- (i) I is a subgroup;
- (ii) if $a \in I$ and $b \in R$, then $ab \in I$.

Usually, an ideal is not a subring. If $1 \in I$ an ideal, then $I = R$.

Lemma 10.2

If $\varphi : R \rightarrow S$ is a homomorphism, then $\ker(\varphi) \triangleleft R$ is an ideal.

Proof. Since φ is a group homomorphism, we know that $\ker(\varphi)$ is a subgroup. If $a \in \ker(\varphi)$ and $b \in R$, then $\varphi(ab) = \varphi(a)\varphi(b) = 0 \cdot \varphi(b) = 0$, so $ab \in \ker(\varphi)$. □

If $u \in R$ is a unit and $I \triangleleft R$ is an ideal, then $u \in I$ implies $I = R$. Since $uv = 1$, we have $1 \in I$ implies $I = R$.

Example 10.1

Let $\mathbb{Z}$ be the ring of integers. Then, $n\mathbb{Z}$ is an ideal. In fact, every ideal has this form for some n . If $I \neq \{0\}$, let $n \in I$ be the smallest positive element that lies in I . We claim that $I = n\mathbb{Z}$. By the Euclidean algorithm, we can write $m = qn + r$ with $0 \leq r < n$ for any $m \in \mathbb{Z}$. We have $qn \in I$ so $r \in I$, but this contradicts the minimality of n .

Definition 10.4 – Principal ideal

Let $A \subseteq R$ be a subset of a ring. The ideal generated by A is

$$(A) = \left\{ \sum_{a \in A} r_a a \mid r_a \in R \text{ with } r_a = 0 \text{ for all but finitely many } a \right\}.$$

An ideal $I \triangleleft R$ is called principal if there exists $r \in R$ such that $I = (r)$.

Example 10.2

The subset $\{f(X) \in R[X] \mid f(0) = 0, \text{ that is it has constant term zero}\} \subseteq R[X]$ is an ideal. It is generated by $X \in R[X]$.

Definition 10.5 – Quotient ring

Let $I \triangleleft R$. The quotient ring is the set of cosets $r + I$ of I in R , with 0 and 1 given by $0_R + I$ and $1_R + I$, and with operations

$$(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I \quad (r_1 + I) \cdot (r_2 + I) = (r_1 r_2) + I$$

Proposition 10.1

The quotient ring R/I is a ring. The function $R \rightarrow R/I$ is a ring homomorphism.
 $r \mapsto r + I$

Proof. This is obviously a well-defined group homomorphism. Multiplication is also well-defined. Suppose $r_1 + I = r'_1 + I$ and $r_2 + I = r'_2 + I$. Then, $r_i - r'_i = a_i$ for some $a_i \in I$. So,

$$r'_1 r'_2 = (r_1 - a_1)(r_2 - a_2) = r_1 r_2 - \underbrace{r_1 a_2 - r_2 a_1 + a_1 a_2}_{\in I}.$$

Hence, $r_1 r_2 + I = r'_1 r'_2 + I$. The rest is straightforward. □

Example 10.3

We have $n\mathbb{Z} \triangleleft \mathbb{Z}$. The quotient ring $\mathbb{Z}/n\mathbb{Z}$ are the cosets $m + n\mathbb{Z}$. In particular, the cosets are $0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \dots, (n-1) + n\mathbb{Z}$. The ring structure is modular arithmetic.

Example 10.4

Consider (X) in $\mathbb{C}[X]$. The elements of the quotient ring $\mathbb{C}[X]/(X)$ can be represented by $(a_0 + a_1 X + \dots + a_n X^n) + (X)$. But the terms $a_1 X, a_2 X^2, \dots, a_n X^n \in (X)$ so the elements can also be represented by $a_0 + (X)$. This representation is unique, and in fact $\mathbb{C}[X]/(X) \cong \mathbb{C}$.

Proposition 10.2 – Euclidean algorithm for polynomials

Let K be a field. If $f, g \in K[X]$, then there exists $q, r \in K[X]$ such that $f = qg + r$ with $\deg(r) < \deg(g)$.

Proof. Let $n = \deg(f)$ so we can write $f = \sum_{i=0}^n a_i X^i$ with $a_i \in K$ and $a_n \neq 0$. Similarly, let $m = \deg(g)$ so we can write $g = \sum_{i=0}^m b_i X^i$ with $b_i \in K$ and $b_m \neq 0$. If $n < m$, use $q = 0$ and $r = f$. If $n \geq m$, we proceed by induction on n . Let $f_1 = f - a_n b_m^{-1} X^{n-m} g$ so $\deg(f_1) < n$. If $n = m$, then $\deg(f_1) < m$ so we can

write $f = (a_n b_m^{-1} X^{n-m})g + f_1$. If $n > m$, by induction, we can write $f_1 = gq_1 + r_1$ with $\deg(r_1) < m$. Then, $f = (a_n b_m^{-1} X^{n-m} + q_1)g + r_1$ and we are done. $\square$

Corollary 10.1

Let K be a field. All ideals in $K[X]$ are principal.

Proof. See Example 10.1 and replace the Euclidean algorithm for $\mathbb{Z}$ with that for $K[X]$. $\square$

We will later see that $\mathbb{Z}[X]$ or $K[X, Y]$ do not have this property.

Theorem 10.1 – First isomorphism theorem

Let $\varphi : R \rightarrow S$ be a ring homomorphism, then the map

$$\begin{aligned} f : R / \ker(\varphi) &\rightarrow \text{im}(\varphi) \\ r + \ker(\varphi) &\mapsto \varphi(r) \end{aligned}$$

is a well-defined ring isomorphism.

Proof. Well-definedness, bijectivity, and the additive homomorphism property all follow from the first isomorphism theorem for groups. We check multiplicativity.

$$f((r + \ker(\varphi)) \cdot (t + \ker(\varphi))) = f(rt + \ker(\varphi)) = \varphi(rt) = \varphi(r)\varphi(t) = f(r\ker(\varphi)) \cdot f(t\ker(\varphi)).$$

$\square$

Example 10.5

Consider the homomorphism

$$\begin{aligned} \varphi : \mathbb{R}[X] &\rightarrow \mathbb{C} \\ f(X) &\mapsto f(i). \end{aligned}$$

This is clearly surjective since $a + bX \mapsto a + bi$. The kernel is exactly the set of polynomials with $f(i) = 0$. Since the coefficients of $f(X) \in \ker(\varphi)$ are real, $-i$ is also a root. So, $X^2 + 1$ divides $f(X) \in \ker(\varphi)$. Hence, the kernel is exactly $(X^2 + 1) \triangleleft \mathbb{R}[X]$. By the first isomorphism theorem, $\mathbb{C} \cong \mathbb{R}[X]/(X^2 + 1)$.

Theorem 10.2 – Second isomorphism theorem

Let $R \leq S$ and $J \triangleleft S$. Then, $R \cap J$ is an ideal in R and $(R + J)/J = \{r + J \mid r \in R\} \leq S/J$ is a subring. Furthermore, $R/(R \cap J) \cong (R + J)/J$.

Proof. Define a function

$$\begin{aligned} \varphi : R &\rightarrow S/J \\ r &\mapsto r + J. \end{aligned}$$

The kernel of φ is $\ker(\varphi) = \{r \in R \mid r + J = 0, \text{ that is } r \in J\} = R \cap J$. The image is $\text{im}(\varphi) = \{r + J \mid r \in R\} = (R + J)/J$. By the first isomorphism theorem, we are done. $\square$

Proposition 10.3 – Correspondence between subrings and ideals

If $I \triangleleft R$, then there is a bijection

$$\begin{aligned}
\{\text{subrings of } R/I\} &\leftrightarrow \{\text{subrings of } R \text{ containing } I\} \\
L \leq R/I &\mapsto \{r \in R \mid r+I \in L\} \\
S/I \leq R/I &\leftarrow I \triangleleft S \leq R.
\end{aligned}$$

Similarly, $\{\text{ideals of } R/I\} \leftrightarrow \{\text{ideals of } R \text{ containing } I\}$.

Theorem 10.3 – Third isomorphism theorem

Let $I \triangleleft R$ and $J \triangleleft R$ with $I \subseteq J$. Then, $J/I \triangleleft R/I$ and $(R/I)/(J/I) \cong R/J$.

Proof. Define a function

$$\begin{aligned}
R/I &\rightarrow R/J \\
r+I &\mapsto r+J.
\end{aligned}$$

This is well-defined by the proof for groups. It is easy to check that this is a ring homomorphism. The kernel is $\{r+I \mid r+J=0\} = J/I$. We then apply the first isomorphism theorem. $\square$

Let R be any ring. There is a unique homomorphism $i : \mathbb{Z} \rightarrow R$ sending 1 to 1_R . Proof as exercise. The kernel of i is $\ker(i) \triangleleft \mathbb{Z}$, so $\ker(i) = n\mathbb{Z}$ for some $n \in \mathbb{Z}$. The number $|n|$ is called the characteristic of R . For example, $\mathbb{Z}/k\mathbb{Z}$ has characteristic k , while $\mathbb{Z}$, $\mathbb{R}$ and $\mathbb{C}$ have characteristic zero.

Also notice that there is no homomorphism from $\{0\}$ to any nonzero ring. This is because $0 = 1$ in the zero ring, but a homomorphism maps 0 to 0 and 1 to 1, so it can't be defined.

11 Integral domains, fields of fractions and prime ideals

Notice that in $\mathbb{Z}/6\mathbb{Z}$, $2 \cdot 3 = 0$ but $2 \neq 0$ and $3 \neq 0$.

Definition 11.1 – Integral domain and zero divisor

A nonzero ring R is an integral domain the following holds for all $a, b \in R$. If $ab = 0$, then either $a = 0$ or $b = 0$. A zero divisor is an element $a \neq 0$ such that there exists $b \neq 0$ with $ab = 0$.

Example 11.1 – Integral domains

Any field is an integral domain. If $ab = 0$ with $b \neq 0$, then $a = abb^{-1} = 0$. Any subring of a field is also an integral domain. Other examples include $\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Q}$, $\mathbb{C}$, $\mathbb{Z}[X]$, $\mathbb{R}[X]$. Notice that products $R \times S$ are never integral domains since $(0_R, 1_S) \cdot (0_S, 1_R) = 0_{R \times S}$.

Lemma 11.1

If R be a finite integral domain, then R is a field.

Proof. Let $a \in R$ be nonzero. The function

$$\begin{aligned}
\mu_a : R &\rightarrow R \\
r &\mapsto ar
\end{aligned}$$

is a group homomorphism. Since R is an integral domain, $\ker(\mu_a) = \{0\}$, so μ_a is injective. Since R is finite, μ_a is also surjective. In particular, there exists a nonzero $b \in R$ such that $ab = 1$. Hence, any nonzero a is a unit, so R is a field. $\square$

Definition 11.2 – Field of fractions

Let R be an integral domain. A field of fractions of R is a field F such that

- (i) $R \leq F$;
- (ii) every $x \in F$ can be written as ab^{-1} where $a, b \in R$ and b^{-1} is the multiplicative inverse of b in F .

Example 11.2

The field $\mathbb{Q}$ is a field of fractions of $\mathbb{Z}$.

Theorem 11.1

Every integral domain R has a field of fractions.

Proof. Define a set $S = \{(a, b) \in R \times R \mid b \neq 0\}$. Let $\sim$ be an equivalence relation on S defined by $(a, b) \sim (c, d)$ if and only if $ad = bc$. We claim that $\sim$ is an equivalence relation. Symmetry and reflexivity is obvious. To check transitivity, suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, then $ad = bc$ and $cf = de$. Multiply the first equality by f and the second by b , so $adf = bde$. Rearranging, $d(af - be) = 0$. Since $d \neq 0$ and R is an integral domain, we have $af - be = 0$ so $(a, b) \sim (e, f)$.

Define $F = S / \sim$. As a notational convenience, write a/b for $[(a, b)]$. Make F into a ring by

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \qquad \frac{a}{c} \cdot \frac{b}{d} = \frac{ab}{cd}$$

Several elementary checks show that F is a ring. To see that F is a field, if $a/b \neq 0_F$, then $a/b \neq 0/1$. So, $a \neq 0$ and we have $b/a \in F$. Moreover, $(a/b)(b/a) = 1$, so F is a field. Finally, define a homomorphism $\varphi : R \rightarrow F$ defined by $\varphi(r) = r/1$. We can check that φ is a ring homomorphism. Observe that $\ker(\varphi) = \{0\}$. By the first isomorphism theorem, $R \cong \text{im}(\varphi) \leq F$. For every $a/b \in F$, we have $a/b = (a/1)(b/1)^{-1}$, so we have completed the proof. $\square$

Proposition 11.1

Let R be a ring. R is a field if and only if the only ideals of R are $\{0\}$ and R .

Proof. If R is a field and $\{0\} \neq I \triangleleft R$, then I contains a unit $u \in I$. But then $1 \in I$ since $1 = uv$. So, $I = R$. Conversely, if the only ideals are $\{0\}$ and R , take $0 \neq r \in R$. Then, (r) is R . So, $1 \in R$ which implies $1 = rs$. So, r is a unit and R is a field. $\square$

Definition 11.3 – Maximal ideal

An ideal $I \triangleleft R$ with $I \neq R$ is called maximal if for any $J \triangleleft R$ such that $I \subseteq J \subseteq R$, we have either $I = J$ or $J = R$.

Note that $\{0\} \triangleleft R$ is maximal if and only if R is a field.

Proposition 11.2

An ideal $I \triangleleft R$ is maximal if and only if R/I is a field.

Proof. By Proposition 11.1, R/I is a field if and only if its only ideals are $\{0\}$ and R/I . We can then apply the correspondence between ideals of R and R/I . $\square$

Definition 11.4 – Prime ideal

An ideal $I \triangleleft R$ with $I \neq R$ is prime if whenever $ab \in I$, either a or b lies in I .

Example 11.3

The ideal $n\mathbb{Z} \triangleleft \mathbb{Z}$ is a prime ideal if and only if n is zero or a prime number. Indeed, if $ab \in p\mathbb{Z}$ for p a prime, then ab is a multiple of p , so either a or b is also a multiple of p . If n is composite, consider $n\mathbb{Z}$. We know that $n = n_1n_2$ for $0 < n_1, n_2 < n$. Then, n_1 and n_2 do not lie in $n\mathbb{Z}$. But their product does so $n\mathbb{Z}$ is not a prime ideal.

Proposition 11.3

An ideal $I \triangleleft R$ is prime if and only if R/I is an integral domain.

Proof. If I is prime, let $(a + I), (b + I) \in R/I$. Suppose $(a + I) \cdot (b + I) = ab + I = 0 + I$. This is true if and only if $ab \in I$. But I is prime, so either a or b lies in I , so either $a + I$ or $b + I$ is $0 + I$ in R/I . Conversely, if R/I is an integral domain, consider $ab \in I$. Then, $0 + I = ab + I = (a + I) \cdot (b + I)$ in R/I . So, either $a + I$ or $b + I$ is equal to $0 + I$. So, either a or b lies in I , hence I is prime. $\square$

Proposition 11.4

Every maximal ideal is a prime ideal.

Proof. Every field is an integral domain. We then apply Proposition 11.2 and Proposition 11.3. $\square$

Note that the converse is false. For example, $\{0\} \triangleleft \mathbb{Z}$.

Proposition 11.5

The characteristic of an integral domain R is either zero or a prime number.

Proof. Consider the unique homomorphism $\mathbb{Z} \rightarrow R$. A characteristic of n implies that $(n) \triangleleft \mathbb{Z}$ is the kernel. By the first isomorphism theorem, $\mathbb{Z}/n\mathbb{Z} \leq R$. Since R is an integral domain, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain. Hence, (n) must be a prime ideal, and n is either zero or prime. $\square$

12 Factorisation in integral domains

Throughout this section, R is an integral domain.

Definition 12.1 – Division

Let $a, b \in R$. We say a divides b , written $a \mid b$, if there exists $c \in R$ such that $b = ac$. Equivalently, $(b) \subseteq (a)$.

Definition 12.2 – Associate

We say $a, b \in R$ are associates if $a = bc$ where $c \in R$ is a unit. Equivalently, $(a) = (b)$. Equivalently, $a \mid b$ and $b \mid a$.

Keep in mind that associates might not be a trivial concept like in the integers. For example, x and $x + x^2 + x^3 + \dots$ are associates in $\mathbb{R}[[X]]$.

Definition 12.3 – Irreducible element and prime element

An element $r \in R$ is irreducible if $r \neq 0$, r is not a unit, and if $r = xy$, then either x or y is a unit.

An element $p \in R$ is prime if $p \neq 0$, p is not a unit, and if $p \mid xy$, then either $p \mid x$ or $p \mid y$.

In $\mathbb{Z}$, the irreducible elements are the prime numbers. The prime elements are also the prime numbers.

Proposition 12.1

Let $0 \neq r \in R$. It is a prime element if and only if (r) is a prime ideal.

Proof. Let (r) be a prime ideal. Prime ideals are not unit by definition, so r is not a unit. Suppose $r \mid ab$, then $ab \in (r)$. But (r) is a prime ideal so either $a \in (r)$ or $b \in (r)$. Hence, $r \mid a$ or $r \mid b$. Conversely, let r be a prime element. Suppose $ab \in (r)$, then $r \mid ab$. But r is a prime element so either $r \mid a$ or $r \mid b$. Hence, $a \in (r)$ or $b \in (r)$. $\square$

Proposition 12.2

If $r \in R$ is a prime element, then r is irreducible.

Proof. Write $r = ab$. Note that $r \mid ab$, so either $r \mid a$ or $r \mid b$. Without loss of generality, assume $r \mid a$. Then, $a = rc$ for some $c \in R$. So, $r = ab = rcb$. We are in an integral domain, so we can cancel r to get $bc = 1$. Hence, b is a unit. $\square$

Example 12.1

Note that the converse is false. Consider the integral domain $R = \mathbb{Z}[\sqrt{-5}] \subseteq \mathbb{C}$. We have a norm function

$$\begin{aligned} N : \mathbb{Z}[\sqrt{-5}] &\rightarrow \mathbb{Z}_{\geq 0} \\ a + b\sqrt{-5} &\mapsto a^2 + 5b^2. \end{aligned}$$

We see this is multiplicative so $N(xy) = N(x)N(y)$. If $1 = uv$, then $N(1) = N(u)N(v)$, so $N(u) = N(v) = 1$ and all units have norm one. Thus, the only units are ± 1 . We claim that $2 \in R$ is irreducible. Observe that $N(2) = 4$. But there is no element $r \in R$ with $N(r) = 2$. So, if $2 = ab$, then $N(a) = 1$ or $N(b) = 1$, and a or b is a unit. Similarly, 3 and $1 \pm \sqrt{-5}$ are all irreducible.

However, all four of these are not prime elements. Observe that $2 \cdot 3 = 6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. We claim that 2 does not divide $1 \pm \sqrt{-5}$. If it did, then $N(2) \mid N(1 \pm \sqrt{-5})$ which is false. We conclude that $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5})$ but 2 does not divide either of them. Hence, 2 is not prime in R .

Definition 12.4 – Euclidean domain

An integral domain R is a Euclidean domain if there exists $\varphi : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$ such that

- (i) $\varphi(ab) \geq \varphi(b)$ for all nonzero a and b ;
- (ii) if $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $\varphi(r) < \varphi(b)$.

$\mathbb{Z}$ with $\varphi(z) = |z|$ is a Euclidean domain. If K is a field, $K[X]$ with $\varphi(p) = \deg(p)$ is also a Euclidean domain. $\mathbb{Z}[\sqrt{-5}]$ is not a Euclidean domain.

Example 12.2

The Gaussian integers $R = \mathbb{Z}[i] \leq \mathbb{C}$ are a Euclidean domain. Define $\varphi(z) = |z|^2$ and we claim that φ makes R into a Euclidean domain. Property (i) is straightforward to check. For (ii), let $a, b \in \mathbb{Z}[i]$ with $b \neq 0$. Consider their ratio $a/b \in \mathbb{C}$. There is a point $q \in \mathbb{Z}[i]$ such that $|a/b - q| < 1$. So we can write $a/b = q + c$ with $|c| < 1$. Then, $a = bq + bc$. Take $r = bc$. We know $r = a - bq \in \mathbb{Z}[i]$ and $\varphi(r) = \varphi(bc) = |bc|^2 < |b|^2 = \varphi(b)$. Hence, property (ii) holds.

In particular, this means that any ideal $I \triangleleft \mathbb{Z}[i]$ is principal.

Definition 12.5 – Principal ideal domain

A principal ideal domain is an integral domain R such that every ideal $I \triangleleft R$ is principal.

Proposition 12.3

Every Euclidean domain is a principal ideal domain.

Proof. Use the same proof as in Example 10.1 but replacing $|\cdot|$ with a Euclidean function φ . □

A non-example. Let $R = \mathbb{Z}[X]$ and $I = (2, X)$. This is not a principal ideal. If $I = (f)$, then $f \mid 2$, so $f \in \{\pm 1, \pm 2\}$. If $f = \pm 1$, then $(f) = \mathbb{Z}[X]$, but $(2, X) \neq \mathbb{Z}[X]$. So $f = \pm 2$, but $2 \nmid X$, thus $(f) \neq (2, X)$. Hence, $(2, X)$ is not principal.

Definition 12.6 – Unique factorisation domain

An integral domain R is a unique factorisation domain such that

- (i) every non-unit in R is a product of irreducibles;
- (ii) if $p_1 p_2 \dots p_n = q_1 q_2 \dots q_m$ with p_i and q_j irreducibles, then $n = m$ and they can be reordered such that p_i and q_i are associates for all i .

Lemma 12.1

In a principal ideal domain R , any irreducible is prime.

Proof. Let $p \in R$ be irreducible. Suppose $p \mid ab$ and $p \nmid a$, we want to show $p \mid b$. Consider $(p, a) \triangleleft R$. This is principal so we can find d such that $(d) = (p, a)$. So, $d \mid a$ and $d \mid p$, and we can write $p = q_1 d$. By the irreducibility of p , either q_1 or d is a unit. If q_1 is a unit, then $d = q_1^{-1} p$. But $d \mid a$ implies $p \mid a$ which contradicts with our assumption. Therefore, d is a unit so $(d) = (p, a) = R$. We can then write $1 = rp + sa$ so $b = rpb + sab$. But p divides the right hand side so $p \mid b$. □

Theorem 12.1

Any principal ideal domain is a unique factorisation domain.

Proof. We make use of the following lemma.

Lemma 12.2

Let R be a principal ideal domain. If $I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \dots$ is a chain of ideals in R , then there exists $N > 0$ such that $I_n = I_{n+1}$ for all $n \geq N$.

Proof. Consider $I = \bigcup_j I_j$. This is an ideal but I is principal so $I = (r)$. So there is some N such that $r \in I_N$. The sequence clearly stabilises here. $\square$

First, we show that $r \in R$ is a product of irreducibles. If r is irreducible, then we are done. Otherwise, write $r = r_1 s_1$ where r_1 and s_1 are not units. If r_1 and s_1 are products of irreducibles, then we are done. Otherwise, without loss of generality, assume r_1 cannot be written as such. Then, we can write $r_1 = r_2 s_2$ where r_2 and s_2 are not units. We continue this process and get that $(r) \subseteq (r_1) \subseteq (r_2) \subseteq \dots$. By Lemma 12.2, this stabilises to say (r_N) . Thus, for $n \geq N$, we have $(r_{n+1} s_{n+1}) = (r_n) = (r_{n+1})$, so s_{n+1} is a unit. Hence, r is a product of irreducibles.

For uniqueness, let $p_1 \dots p_n = q_1 \dots q_m$ be a product of irreducibles, so $p_1 \mid q_1 \dots q_m$. By Lemma 12.1, p_1 is prime, so $p_1 \mid q_i$ for some i . But q_i is a irreducible, so $q_i = ap_1$ with a being a unit. Relabelling q_i as q_1 and vice versa, we have $p_1 \dots p_n = aq_1 \dots q_m$ so $p_2 \dots p_n = aq_2 \dots q_m$. Replace aq_2 with q_2 and repeat the process. Thus, p_i and q_i are associates for all i . This also shows that $n = m$. If $n = m + k$, then we would get that $\prod_{i=m+1}^n p_i$ is a unit, which is false. $\square$

Note that the converse is false. Also, in a unique factorisation domain, any irreducible is prime.

Definition 12.7 – Greatest common divisor and least common multiple

Let $a_1, \dots, a_n \in R$. We say d is a greatest common divisor (gcd) if $d \mid a_i$ for all i , and if any other d' satisfies $d' \mid a_i$ for all i , then $d' \mid d$.

Similarly, we say m is a least common multiple (lcm) if $a_i \mid m$ for all i , and if any other m' satisfies $m' \mid a_i$ for all i , then $m \mid m'$.

Theorem 12.2

Let R be a unique factorisation domain, then gcd's and lcm's exist, and are unique up to associates.

Proof. Let $p_1, \dots, p_m$ be the list of all the irreducible factors in the factorisations of all the a_i , with no two of them associates. Write $a_i = u_i \prod_{j=1}^m p_j^{n_{ij}}$ with u_i a unit and $n_{ij} \geq 0$. Let $m_j = \min_i \{n_{ij}\}$ and choose $d = \prod_{j=1}^m p_j^{m_j}$. By definition, $m_j \leq n_{ij}$ so we know that $d \mid a_i$. If $d' \mid a_i$ for all i , then $d' = v \prod_{j=1}^m p_j^{t_j}$ with v a unit. So, we must have $t_j \leq m_j$ for all j . Hence, $d' \mid d$. Uniqueness is immediate, if $d' \mid d$ and $d \mid d'$, then d and d' are associates. The argument for lcm is analogous. $\square$

13 Factorisation in polynomial rings

If K is a field, then $K[X]$ is a Euclidean domain. We know the following.

- (i) Any ideal $I \triangleleft K[X]$ is principal, so $I = (f)$.
- (ii) $f \in K[X]$ is irreducible if and only if it is prime.
- (iii) Let $f \in K[X]$ be irreducible. Consider $(f) \triangleleft K[X]$. Suppose $(f) \subseteq J \triangleleft K[X]$ and $J = (g)$. Then, $f = gh$. But either g and f are associates, or g is a unit. Hence, (f) must be maximal. In particular, f is irreducible if and only if $K[X]/(f)$ is a field.

Definition 13.1 – Content and primitive polynomial

Let R be a unique factorisation domain and $f = a_0 + a_1X + \cdots + a_nX^n \in R[X]$. The content of f is $c(f) = \gcd(a_0, a_1, \dots, a_n) \in R$. If $c(f)$ is a unit, we say that f is primitive.

We will work towards the following theorem.

Theorem 13.1 – Gauss's lemma

Let R be a unique factorisation domain and $f \in R[X]$. If f is primitive, then f is reducible in $R[X]$ if and only if f is reducible in $F[X]$, where F is the field of fractions of R .

The requirement that f is primitive is necessary. For example, $2X + 2 \in \mathbb{Z}[X]$ is not primitive. It is reducible in $\mathbb{Z}[X]$ but is irreducible in $\mathbb{Q}[X]$.

Example 13.1

Consider $f = X^3 + X + 1 \in \mathbb{Z}[X]$. $c(f) = 1$ so Gauss's lemma holds. If f were reducible, we can write $X^3 + X + 1 = gh$ with $g, h \in \mathbb{Z}[X]$ not units. They cannot have degree zero since we are working in $\mathbb{Z}[X]$. So, we can let $g = b_0 + b_1X$ and $h = c_0 + c_1X + c_2X^2$. Substituting in, b_0 and b_1 are ± 1 , but ± 1 is not a root of f . Thus, f is irreducible in $\mathbb{Z}[X]$ and hence irreducible in $\mathbb{Q}[X]$.

Lemma 13.1

Let R be a unique factorisation domain. If $f, g \in R[X]$ are primitive, then $fg \in R[X]$ is primitive.

Proof. Let $f = a_0 + a_1x + \cdots + a_nx^n$ and $g = b_0 + b_1x + \cdots + b_mx^m$ with $a_n, b_m \neq 0$ and f, g primitive. Suppose $c(fg)$ is not a unit, then we can find $p \in R$ irreducible such that $p \mid c(fg)$. Since f and g are primitive, $p \nmid c(f)$ and $p \nmid c(g)$. Suppose k and l are the smallest indices such that $p \nmid a_k$ and $p \nmid b_l$. Consider the coefficient of x^{k+l} , which is $\sum_{i+j=k+l} a_i b_j$. By our assumption, this sum is divisible by p . Also, p divides $a_{k-1}b_{l+1}, \dots, a_0b_{l+k}$ and $a_{k+1}b_{l-1}, \dots, a_{l+k}b_0$. Therefore, p divides $a_k b_l$. But p is prime as R is a UFD, so $p \mid a_k$ or $p \mid b_l$. This gives a contradiction. $\square$

Corollary 13.1

Let R be a unique factorisation domain and $f, g \in R[X]$. Then, $c(fg)$ and $c(f)c(g)$ are associates.

Proof. Write $f = c(f)f_1$ and $g = c(g)g_1$ with f_1 and g_1 primitive. Then, $fg = c(f)c(g)f_1g_1$ and f_1g_1 is primitive by Lemma 13.1. Thus, $c(f)c(g)$ is a gcd for the coefficients of fg . This gives the result. $\square$

We can now prove Gauss's lemma.

Proof. Let $f = gh$ in $R[X]$ with g and h non-units. Since f is primitive, g and h are both primitive, so $\deg(g), \deg(h) > 0$. Hence, it is reducible in $F[X]$.

Conversely, let $f = gh$ in $F[X]$ with g and h non-units, so $\deg(g), \deg(h) > 0$. Find a and b such that

$abf = (ag)(bh)$ with $(ag), (bh) \in R[X]$. Write $(ag) = c(ag)g_1$ and $(bh) = c(bh)h_1$ with g_1 and h_1 primitive. Since f is primitive, $c(abf) = ab$. By Corollary 13.1, $ab = c(abf) = c((ag)(bh)) = c(ag)c(bh)u$ where u is a unit. Also, $abf = c(ag)c(bh)g_1h_1 = u^{-1}abg_1h_1$. We can cancel ab to get $f = u^{-1}g_1h_1$ and we are done. $\square$

Proposition 13.1

Let R be a unique factorisation domain and F be its field of fractions. Let $g \in R[X]$ be primitive. If $f \in R[X]$ is divisible by g in $F[X]$, then f is also divisible by g in $R[X]$. In other words, letting $I = (g) \triangleleft R[X]$ and $J = (g) \triangleleft F[X]$, then $I = J \cap R[X]$.

Proof. Let $f \in J \cap R[X]$. We can write $f = gh$ in $F[X]$. Now multiply by $b \in R$ to get $bf = g(bh)$ in $R[X]$. Set $(bh) = c(bh)h_1$ with h_1 primitive. Since gh_1 is primitive, $c(bh) = c(bf)u$ where u is a unit. Since $b, f \in R[X]$, we have $c(bf) = bc(f)$. Now observe that $bf = gubc(f)h_1$ and we can cancel b to get divisibility. $\square$

Theorem 13.2

If R is a unique factorisation domain, then $R[X]$ is a unique factorisation domain.

Proof. Let $f \in R[X]$. Write $f = c(f)f_1$ with f_1 primitive. Factorise $c(f)$ as $c(f) = p_1 \dots p_n$ with p_i irreducible. We now produce a factorisation of f_1 by induction on its degree. If f_1 is irreducible, then we are done. Otherwise, write $f_1 = f_{11}f_{12}$ with f_{11} and f_{12} having smaller degrees and necessarily primitive. We continue for f_{11} and f_{12} . The process terminates because the degree drops at each step. So, we can write $f_1 = q_1 \dots q_m$ with q_i irreducible. Thus, $f = p_1 \dots p_n q_1 \dots q_m$.

The uniqueness for the p_i 's follows immediately from R being a UFD. Uniqueness for the q_i 's is as follows. Consider $f_1 = q_1 \dots q_m = r_1 \dots r_l$ in $R[X]$. We can view this in $F[X]$. But $F[X]$ is a Euclidean domain and hence a UFD. So, $l = m$ and up to reordering, the q_i 's and the r_i 's are associates. By Proposition 13.1, they are also associates in $R[X]$. $\square$

Examples of unique factorisation domains include $\mathbb{Z}[X]$, $\mathbb{R}[X, Y]$, $F[X, Y]$, $\mathbb{Z}[X_1, \dots, X_n]$ and so on.

Theorem 13.3 – Eisenstein's criterion

Let R be a unique factorisation domain and $f = a_0 + a_1X + \dots + a_nX^n \in R[X]$ with $a_n \neq 0$. Let p be irreducible such that

- (i) $p \nmid a_n$;
- (ii) $p \mid a_i$ for all $0 \leq i < n$;
- (iii) $p^2 \nmid a_0$.

Then, f is irreducible in $R[X]$ and equivalently in $F[X]$ where F is the field of fractions of R .

Proof. Suppose $f = gh$. Write $g = r_0 + r_1X + \dots + r_kX^k$ and $h = s_0 + s_1X + \dots + s_lX^l$ with $r_k \neq 0$ and $s_l \neq 0$. By our hypothesis, $p \nmid r_k$ and $p \nmid s_l$. Also, p divides either r_0 or s_0 but not both, say $p \mid r_0$. Let j be such that $p \mid r_0, p \mid r_1, \dots, p \mid r_{j-1}, p \nmid r_j$. Note that $a_j = r_0s_j + r_1s_{j-1} + \dots + r_js_0$. This shows that $p \nmid a_j$ because $p \nmid s_0$ and $p \nmid r_j$. Therefore, we must have $j = n$ and so $k = n$. Thus, $\deg(g) = n$ and $\deg(h) = 0$. Hence, $f = gh$ is not a proper factorisation since h is a unit. $\square$

Example 13.2

Consider $R = \mathbb{Z}$ and $f = X^n - p$ with p prime. This is irreducible by Eisenstein's criterion.

Example 13.3

Let p be a prime and $f = X^{p-1} + X^{p-2} + \cdots + X + 1 \in \mathbb{Z}[X]$. We cannot satisfy Eisenstein's criterion for f , but we can apply it to $\hat{f}(Y) = f(Y+1)$. We can calculate $\hat{f}(Y) = Y^{p-1} + \binom{p}{1}Y^{p-2} + \cdots + \binom{p}{p-1}$ so Eisenstein's criterion does imply that $\hat{f}(Y)$ is irreducible. This means that $f(X)$ is irreducible. If it were with $f(X) = f_1 f_2$, then we can do the substitution $Y = X - 1$ to get a factorisation of $\hat{f}(Y)$.

14 Gaussian integers

The ring of Gaussian integers is $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$. It has a norm $N(a + bi) = a^2 + b^2$. This norm enables division with remainder so $\mathbb{Z}[i]$ is a Euclidean domain, which is a principal ideal domain, which is a unique factorisation domain.

- (i) Irreducibles coincides with the primes in $\mathbb{Z}[i]$.
- (ii) The units have norm 1, and are exactly ± 1 and $\pm i$.
- (iii) We have $2 = (1 + i)(1 - i)$ so 2 is not a prime. 3 is a prime, if $3 = uv$ with $u, v \in \mathbb{Z}[i]$, then $N(3) = 9 = N(u)N(v)$. Either $N(u)$ or $N(v)$ is a unit, or $N(u) = N(v) = 3$. But $u^2 + v^2 = 3$ has no integer solutions, so either u or v is a unit. We see that the primes in $\mathbb{Z}[i]$ are distinctly different from the primes in $\mathbb{Z}$.

Proposition 14.1

A prime $p \in \mathbb{Z}$ remains prime in $\mathbb{Z}[i]$ if and only if $p \neq a^2 + b^2$ with $a, b \in \mathbb{Z} \setminus \{0\}$.

Proof. If $p = a^2 + b^2$, then $p = (a + bi)(a - bi)$ and p is not irreducible. Conversely, suppose $p = uv$ with u and v not units in $\mathbb{Z}[i]$. Then, $p^2 = N(p) = N(u)N(v)$, so $N(u) = p$. Write $u = a + bi$, so $p = a^2 + b^2$. $\square$

Theorem 14.1

The primes in $\mathbb{Z}[i]$ up to associates are

- (i) the primes $p \in \mathbb{Z}$ such that $p \equiv 3 \pmod{4}$;
- (ii) Gaussian $z \in \mathbb{Z}[i]$ with $N(z) = p$ for some prime p such that $p = 2$ or $p \equiv 1 \pmod{4}$.

Proof. We require the following lemma.

Lemma 14.1

Let p be a prime and $\mathbb{F}_p$ be the field with p elements. Then, the group $\mathbb{F}_p^\times$ of nonzero elements under multiplication is cyclic, so $\mathbb{F}_p^\times \cong C_{p-1}$.

Proof. $\mathbb{F}_p^\times$ is abelian with order $p - 1$. By the classification of finite abelian groups, if $\mathbb{F}_p^\times$ is not cyclic, then it contains $C_m \times C_m$ with $m > 1$ as a subgroup. This means there are at least m^2 elements in $\mathbb{F}_p$ such that $X^m - 1 = 0$. But $X^m - 1$ can have at most m roots over a field. $\square$

Proof. We first show these are primes. If $p \equiv 3 \pmod{4}$, then $p \neq a^2 + b^2$, since a square number mod 4 is always 0 or 1. So these are primes in $\mathbb{Z}[i]$.

On the other hand, if $N(z) = p$, and $z = uv$, then $N(u)N(v) = p$. So $N(u)$ is 1 or $N(v)$ is 1. So u or v is a unit. Note that we did not use the condition that $p \not\equiv 3 \pmod{4}$. This is not needed, since $N(z)$ is always a sum of squares, and hence $N(z)$ cannot be a prime that is 3 mod 4.

Now let $z \in \mathbb{Z}[i]$ be irreducible, hence prime. Then $\bar{z}$ is also irreducible. So $N(z) = z\bar{z}$ is a factorization of $N(z)$ into irreducibles. Let $p \in \mathbb{Z}$ be a prime number dividing $N(z)$, which exists since $N(z) \neq 1$.

Now if $p \equiv 3 \pmod{4}$, then p itself is prime in $\mathbb{Z}[i]$ by the first part of the proof. So $p \mid N(z) = z\bar{z}$. So $p \mid z$ or $p \mid \bar{z}$. Note that if $p \mid \bar{z}$, then $p \mid z$ by taking complex conjugates. So we get $p \mid z$. Since both p and z are both irreducible, they must be equal up to associates.

Otherwise, we get $p = 2$ or $p \equiv 1 \pmod{4}$. If $p \equiv 1 \pmod{4}$, then $p - 1 = 4k$ for some $k \in \mathbb{Z}$. As $\mathbb{F}_p^\times \cong C_{p-1} = C_{4k}$, there is a unique element of order 2 (this is true for any cyclic group of even order), which must be $[-1] \in \mathbb{F}_p$. Now let $a \in \mathbb{F}_p^\times$ be an element of order 4. Then a^2 has order 2, so $[a^2] = [-1]$.

This is a complicated way of saying we can find an a such that $p \mid a^2 + 1$. Thus $p \mid (a+i)(a-i)$. In the case where $p = 2$, we know by checking directly that $2 = (1+i)(1-i)$.

In either case, we deduce that p is not prime (hence irreducible), since it clearly does not divide $a \pm i$ (or $1 \pm i$). So we can write $p = z_1 z_2$, for $z_1, z_2 \in \mathbb{Z}[i]$ not units. Now we get

$$p^2 = N(p) = N(z_1)N(z_2).$$

As the z_i are not units, we know $N(z_1) = N(z_2) = p$. By definition, this means $p = z_1 \bar{z}_1 = z_2 \bar{z}_2$. But also $p = z_1 z_2$. So we must have $\bar{z}_1 = z_2$.

Finally, we have $p = z_1 \bar{z}_1 \mid N(z) = z\bar{z}$. All these z, z_i are irreducible. So z must be an associate of z_1 (or maybe $\bar{z}_1$). So in particular $N(z) = p$. □

□

The proof gives a way to find $\alpha \in \mathbb{Z}[i]$ such that $N(\alpha) = p$ whenever $p \equiv 1 \pmod{4}$.

Corollary 14.1

An integer n can be written as a sum of squares $x^2 + y^2$ if and only if when we write $n = p_1^{n_1} \dots p_k^{n_k}$ as a product of distinct primes, then $p_i \equiv 3 \pmod{4}$ implies n_i is even.

Proof. If $n = x^2 + y^2$, then $n = N(x + iy)$. Let $z = x + iy$ and factorise it $z = \alpha_1 \dots \alpha_q$ into irreducibles. By Theorem 14.1, either α_i is a prime of $\mathbb{Z}$ that is 3 mod 4, or $N(\alpha_i)$ is a prime that is 2 or 1 mod 4. We take norm to get $x^2 + y^2 = N(z) = \prod N(\alpha_i)$, and each $N(\alpha_i) = p^2$ for $p \equiv 3 \pmod{4}$, or $N(\alpha_i) = p$ for $p = 2$ or $p \equiv 1 \pmod{4}$. This gives one implication.

Conversely, let $n = \prod p_i^{n_i}$ with the given conditions. In this case, if $p_i \equiv 3 \pmod{4}$, then $p_i^{n_i} = (p_i^{n_i/2})^2 = N(p_i^{n_i/2})$. If $p = 2$ or $p \equiv 1 \pmod{4}$, then we know $p_i = N(\alpha_i)$ from the proof of Theorem 14.1. Since the norm is multiplicative, we can write $N(n) = N(z)$ for some $z \in \mathbb{Z}[i]$. Hence, we can write $z = x^2 + y^2$. □

Example 14.1

The above gives a way to write 65 as a sum of squares. Note that $65 = 13 \times 5$. We have $13 = (2 + 3i)(2 - 3i)$ and $5 = (2 + i)(2 - i)$. Pick one factor from each and we get $65 = N((2 + 3i)(2 + i)) = N(1 + 8i) = 1^2 + 8^2$. Other factors give $65 = 7^2 + 4^2$.

15 Algebraic integers

Definition 15.1 – Algebraic integer

A complex number $\alpha \in \mathbb{C}$ is an algebraic integer if it is the root of a monic polynomial in $\mathbb{Z}[X]$.

If α is an algebraic integer, we write $\mathbb{Z}[\alpha]$ for the smallest subring of $\mathbb{C}$ containing α , so $\mathbb{Z}[\alpha] = \bigcap_{\alpha \in R \leq \mathbb{C}} R$. Equivalently, it is the image of the evaluation map $e_\alpha : \mathbb{Z}[X] \rightarrow \mathbb{C}$

$$g(X) \mapsto g(\alpha).$$

Proposition 15.1

Let α be an algebraic integer. The ideal $I = \ker(e_\alpha) \triangleleft \mathbb{Z}[X]$ is principal. It is equal to (f_α) with f_α irreducible and monic.

Proof. By definition of an algebraic integer, we know $I = \ker(e_\alpha)$ is non-zero. Take $f_\alpha \in I$ to be of minimal degree. We can assume f_α is primitive because α satisfies a monic polynomial. We hope to show $I = (f_\alpha)$. Let $h \in I$. Apply the Euclidean algorithm in $\mathbb{Q}[X]$ to get $h = f_\alpha q + r$ where $\deg(r) < \deg(f_\alpha)$. We clear the denominators by multiplying by some big $a \in \mathbb{Z}$, so $ah = af_\alpha q + ar$. Plug in α , we have $(ah)(\alpha) = 0$ and $f_\alpha(\alpha) = 0$, so $(ar)(\alpha) = 0$. But by the minimality of degree of f_α , we get $ar = 0$, so $r = 0$. Therefore, $ah = af_\alpha q$. Since f_α is primitive, $c(ah) = c(aq)$. Since $h \in \mathbb{Z}[X]$, a divides $c(ah)$, and so a divides $c(aq)$. Thus, we can write $aq = a\bar{q}$ with $\bar{q} \in \mathbb{Z}[X]$. Cancel the a 's to get $q = \bar{q}$. Hence, f_α divides h , and so $I = (f_\alpha)$.

To see that it is irreducible, observe that $\mathbb{Z}[X]/\ker(e_\alpha) \cong \mathbb{Z}[X]/(f_\alpha) \leq \mathbb{C}$, so $\mathbb{Z}[X]/(f_\alpha)$ is an integral domain since $\mathbb{C}$ is an integral domain. So, f_α is prime and hence irreducible. To see this, suppose $f_\alpha = pq$ for some non-units p and q . Since $f_\alpha(\alpha) = 0$, $p(\alpha)q(\alpha) = 0$. Since $p(\alpha), q(\alpha) \in \mathbb{C}$ which is an integral domain, we must have say $p(\alpha) = 0$. But then $\deg(p) < \deg(f_\alpha)$ which is a contradiction. $\square$

Proposition 15.2

Let $\alpha \in \mathbb{C}$ be an algebraic integer. If $\alpha \in \mathbb{Q}$, then $\alpha \in \mathbb{Z}$.

Proof. Let f_α be the polynomial generating $\ker(e_\alpha)$. This is irreducible in $\mathbb{Z}[X]$. By Gauss's lemma, it is also irreducible in $\mathbb{Q}[X]$. But $(X - \alpha)$ divides f_α in $\mathbb{Q}[X]$, so $f_\alpha = X - \alpha \in \mathbb{Z}[X]$. $\square$

16 Hilbert basis theorem

Definition 16.1 – Noetherian ring

A ring R is Noetherian if every chain $I_1 \subseteq I_2 \subseteq \dots$ of ideals in R stabilises. That is, there exists $N > 0$ such that $I_{n+1} = I_n$ for all $n \geq N$. This is known as the ascending chain condition.

Definition 16.2 – Finite generated ideal

An ideal is finitely generated if it is equal to $(r_1, \dots, r_k)$ for some $r_i \in R$.

Proposition 16.1

A ring R is Noetherian if and only if all ideals are finitely generated.

Proof. Suppose all the ideals are finitely generated. Given $I_1 \subseteq I_2 \subseteq \dots$ a chain of ideals, consider $I = \bigcup_{n \geq 1} I_n$. This is an ideal, and so is finitely generated by say $r_1, \dots, r_k$. But there is an N such that $r_i \in I_N$ for all i , hence $I_{j+1} = I_j$ for all $j \geq N$.

Conversely, if there is an ideal I not finitely generated, pick $r_1 \in I$ such that $(r_1) \neq I$. Pick $r_2 \in I \setminus (r_1)$ and so on. Then, $(r_1) \subsetneq (r_1, r_2) \subsetneq (r_1, r_2, r_3) \subsetneq \dots$ does not satisfy the ascending chain condition. $\square$

Theorem 16.1 – Hilbert basis theorem

If R is Noetherian, then $R[X]$ is Noetherian.

Proof. Let $J \triangleleft R[X]$. Let $f_1 \in J$ have minimal degree. If $J = (f_1)$, then we are done. If not, then choose $f_2 \in J \setminus (f_1)$ of minimal degree. We continue similarly to produce a sequence of f_i 's. Let $a_i \in R$ be the leading coefficient of f_i . The chain of ideals $(a_1) \subseteq (a_1, a_2) \subseteq (a_1, a_2, a_3) \subseteq \dots$ stabilises since R is Noetherian. Thus, $(a_1, a_2, \dots) = (a_1, \dots, a_m)$ for some m . Let $a_{m+1} = \sum_{i=1}^m a_i b_i$ with $b_i \in R$.

Consider the polynomial $g = \sum_{i=1}^m b_i f_i X^{\deg(f_{m+1}) - \deg(f_i)}$. Note that $\deg(f_{m+1}) \geq \deg(f_i)$ for all $i \leq m$ so g is in $R[X]$. Observe that g has the same degree as f_{m+1} and the same leading coefficient as f_{m+1} . Then $f_{m+1} - g$ has a smaller degree than f_{m+1} but it is not in $(f_1, \dots, f_m)$. This contradicts the minimality of $\deg(f_{m+1})$. Hence, the process of choosing the f_i 's must terminate, and $R[X]$ is Noetherian by Proposition 16.1. $\square$

This implies that rings like $\mathbb{Z}[X]$, $\mathbb{C}[X]$ and $\mathbb{C}[X_1, X_2, X_3]$ are all Noetherian.

Proposition 16.2

If R is Noetherian and $I \triangleleft R$, then R/I is Noetherian.

Proof. Let $J \triangleleft R/I$. By Proposition 10.3, let $J' \triangleleft R$ be the corresponding ideal. Since R is Noetherian, J' is generated by some $r_1, \dots, r_k$. Since J is the image of J' under $R \rightarrow R/I$, J is generated by $r_1 + I, \dots, r_k + I$. $\square$

Principal ideal domains are Noetherian. The converse is false, for example, $\mathbb{C}[X, Y]/(Y^2 - X^2)$ is Noetherian but not a principal ideal domain.

Note that subrings of Noetherian rings need not be Noetherian. Let $R = \mathbb{C}[X_1, X_2, \dots]$. Let F be the fraction field of R . $R \leq F$ is not Noetherian but F is Noetherian since F is a field.

17 Modules

Definition 17.1 – R -module

Let R be a ring and $(M, +, 0)$ be an abelian group. A quadruple $(M, +, 0, \cdot)$ is an R -module if the operation $\cdot : R \times M \rightarrow M$ satisfies the following for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$.

$$(r, m) \mapsto rm$$

- (i) $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$;
- (ii) $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$;
- (iii) $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$;
- (iv) $1_R \cdot m = m$.

Example 17.1

If $R = K$ is a field, then an R -module M is the same thing as a K -vector space. Any abelian group G is a $\mathbb{Z}$ -module by $n \cdot g := g + \cdots + g$ (n times) when n is positive and $n \cdot g := (-g) + \cdots + (-g)$ ($-n$ times) when n is negative.

Example 17.2

For any ring R , the product $R^m = R \times \cdots \times R$ is an R -module via

$$\begin{aligned} R \times R^m &\rightarrow R^m \\ (r, (r_1, \dots, r_m)) &\mapsto (rr_1, \dots, rr_m). \end{aligned}$$

If $I \triangleleft R$ is an ideal, then I is an R -module via

$$\begin{aligned} R \times I &\rightarrow I \\ (r, x) &\mapsto rx. \end{aligned}$$

Similarly, R/I is also an R -module via

$$\begin{aligned} R \times R/I &\rightarrow R/I \\ (r, x + I) &\mapsto rx + I. \end{aligned}$$

Example 17.3

Let V be a K -vector space and α be an endomorphism of V (a linear transformation from V to itself). We can make V into a $K[X]$ -module via

$$\begin{aligned} K[X] \times V &\rightarrow V \\ (f(X), v) &\mapsto (f(\alpha))(v). \end{aligned}$$

In other words, $(X, v) \mapsto \alpha(v)$. This very much depends on the choice of α .

Example 17.4

Take a ring homomorphism $\varphi : R \rightarrow S$. This makes S an R -module by

$$\begin{aligned} R \times S &\rightarrow S \\ (r, s) &\mapsto \varphi(r)s. \end{aligned}$$

For example, the ring homomorphism $\mathbb{C}[X] \rightarrow \mathbb{C}[X]/(X^2)$ makes $\mathbb{C}[X]/(X^2)$ a $\mathbb{C}[X]$ -module. Similarly, $\mathbb{C}[X, Y]$ is a $\mathbb{C}[X]$ -module by the inclusion map.

Definition 17.2 – Submodule

Let M be an R -module. A subgroup N of M is a submodule if $rn \in N$ for any $n \in N$ and $r \in R$. We denote this by $N \leq M$.

Let $R = M = \mathbb{Q}$, but $N = \mathbb{Z} \subseteq \mathbb{Q}$ is not a $\mathbb{Q}$ -submodule even though it is a subgroup. If R is a field, then an R -submodule is the same as a vector subspace.

Definition 17.3 – Quotient module

Let $N \leq M$ be an R -submodule. The quotient R -module is the abelian group M/N with R -module structure

$$\begin{aligned} R \times M/N &\rightarrow M/N \\ (r, m + N) &\mapsto rm + N. \end{aligned}$$

It is easy to check that this is well-defined and gives a R -module.

Definition 17.4 – R -module homomorphism

Let M and N be R -modules. An R -module homomorphism is a group homomorphism $\varphi : M \rightarrow N$ such that

$$\varphi(rm) = r\varphi(m)$$

for all $r \in R$ and $m \in M$. A bijective R -module homomorphism is called an R -module isomorphism.

Theorem 17.1 – First isomorphism theorem

Let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Then, $\ker(\varphi) = \{m \in M \mid \varphi(m) = 0\}$ is an R -submodule of M . Similarly, $\text{im}(\varphi) = \{\varphi(m) \mid m \in M\}$ is an R -submodule of N . There is an isomorphism

$$\begin{aligned} M/\ker(\varphi) &\rightarrow \text{im}(\varphi) \\ m + \ker(\varphi) &\mapsto \varphi(m). \end{aligned}$$

Theorem 17.2 – Second isomorphism theorem

Let $K, L \leq M$ be R -submodules. Then, $K + L = \{k + l \mid k \in K, l \in L\}$ is an R -submodule of M . There is an isomorphism $(K + L)/K \cong L/(K \cap L)$.

Theorem 17.3 – Third isomorphism theorem

Let $N \leq L \leq M$ be R -submodules. Then, $M/L \cong (M/N)/(L/N)$.

Proposition 17.1

Let $N \leq M$ be an R -submodule. There is a correspondence

$$\{\text{submodules of } M/N\} \leftrightarrow \{\text{submodules of } M \text{ that contain } N\}.$$

Definition 17.5 – Annihilator

Let M be an R -module. For $m \in M$, its annihilator is

$$\text{Ann}(m) = \{r \in R \mid rm = 0\}.$$

More generally, for $S \subseteq M$, its annihilator is

$$\text{Ann}(S) = \{r \in R \mid rs = 0 \text{ for all } s \in S\}.$$

In particular, $\text{Ann}(M) = \bigcap_{m \in M} \text{Ann}(m)$.

The annihilator $\text{Ann}(S)$ is an ideal in R . In the theory of vector spaces, the annihilators are trivial. There are more interesting in general R -modules. For example, when $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$.

Definition 17.6 – Submodule generated

Given an element $m \in M$, the submodule generated by m is $Rm = \{rm \mid r \in R\} \leq M$.

Proposition 17.2

For $m \in M$, there is an isomorphism $R / \text{Ann}(m) \cong Rm$.

Proof. Fix $m \in M$. Consider the function $R \rightarrow M$ defined by $r \mapsto rm$. This is an R -module homomorphism. Applying the first isomorphism theorem gives the result. $\square$

Definition 17.7 – Finitely generated module

An R -module M is finitely generated if there exist $m_1, \dots, m_k \in M$ such that

$$M = Rm_1 + \dots + Rm_k = \{r_1m_1 + \dots + r_km_k \mid r_i \in R\}.$$

This is analogue to the notion of a vector space being finite dimensional.

Lemma 17.1

An R -module M is finitely generated if and only if there exists a surjective R -module homomorphism $R^k \rightarrow M$ for some k .

Proof. If M is finitely generated, then $M = Rm_1 + \dots + Rm_k$ for some $m_1, \dots, m_k \in M$. Set

$$\begin{aligned} \varphi : R^k &\rightarrow M \\ (r_1, \dots, r_k) &\mapsto r_1m_1 + \dots + r_km_k \end{aligned}$$

which is clearly surjective. Conversely, given a surjective $\varphi : R^k \rightarrow M$, let $e_i = (0, \dots, 0, 1, 0, \dots, 0) \in R^k$ where the 1 is in the i -th position. Define $m_i = \varphi(e_i)$. It is easy to check that $M = Rm_1 + \dots + Rm_k$. $\square$

Corollary 17.1

Let M be finitely generated. If $N \leq M$ is a submodule, then M/N is finitely generated.

Proof. By Lemma 17.1, consider the surjective $R^k \rightarrow M$. The composition $R^k \rightarrow M \rightarrow M/N$ is also surjective, so we are done. $\square$

Note that submodules do not have this property. Take $R = \mathbb{C}[X_1, X_2, \dots]$ and $M = R$. This is finitely generated since $R \rightarrow M$ is surjective. Take $I = (X_1, X_2, \dots) \leq M$. This is not finitely generated.

18 Direct sums and free modules

Definition 18.1 – Direct sum

Let $M_1, \dots, M_k$ be R -modules. The direct sum $M_1 \oplus \dots \oplus M_k$ is the group $M_1 \times \dots \times M_k$ with R -action $r(m_1, \dots, m_k) = (rm_1, \dots, rm_k)$.

Definition 18.2 – Linear independence

Let $m_1, \dots, m_k \in M$. The set $\{m_1, \dots, m_k\}$ is linearly independent if whenever $\sum_i r_i m_i = 0$ for some $r_i \in R$, we have $r_i = 0$ for all i .

Definition 18.3 – Freely generating set and basis

A subset $S \subseteq M$ freely generates M if the following hold.

- (i) S generates M ;
- (ii) Any set function $f : S \rightarrow N$ to another R -module N extends to an R -module homomorphism $\varphi_f : M \rightarrow N$ such that $\varphi_f(s) = f(s)$ for all $s \in S$.

A module is free if it is freely generated by $S \subseteq M$. We call such S a basis.

Note that free generating sets may not exist. Consider the $\mathbb{Z}$ -module $\mathbb{Z}/2\mathbb{Z}$. If it were freely generated, the only possibility for a basis is $S = \{1\}$. But this is not free because if we set $N = \mathbb{Z}$ and $f(1) = 1$, we would need a nonzero $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}$, which does not exist. An actual example of a free $\mathbb{Z}$ -module is $\mathbb{Z}^k$.

Proposition 18.1

Let $S = \{m_1, \dots, m_k\} \subseteq M$. The following are equivalent.

- (i) S generates M freely.
- (ii) S generates M and S is linearly independent.
- (iii) Every $m \in M$ can be uniquely expressed as $m = \sum_{i=1}^k r_i m_i$.

Proof. The fact that (ii) and (iii) are equivalent is familiar from linear algebra, with an identical proof. For (i) implies (ii), let S generate M freely. Suppose S is not linearly independent, we can write $r_1 m_1 + \dots + r_k m_k = 0$ with $r_1 \neq 0$ by relabelling. Define $f : S \rightarrow R$ by $m_1 \mapsto 1$ and $m_i \mapsto 0$ for all $i > 1$. This extends to $\varphi_f : M \rightarrow R$. But $0 = \varphi_f(0) = \varphi_f(\sum r_i m_i) = r_1 \neq 0$, which is a contradiction.

Conversely, we prove (iii) implies (i). If any $m \in M$ can be uniquely expressed as $m = \sum r_i m_i$, then given any set function $f : S \rightarrow N$, we can define $\varphi_f(m) = \sum r_i f(m_i)$. This is well-defined by the unique expression of m . This is a homomorphism, so it follows that S freely generates M . $\square$

Definition 18.4 – Module of relations and finite presented

Let M be finitely generated and $\theta : R^k \rightarrow M$ be the surjective homomorphism discussed in Lemma 17.1. Then, the kernel of θ is called the module of relations for θ . We say M is finitely presented if $\ker(\theta)$ is finitely generated.

Proposition 18.2

Let R be a nonzero ring. If $R^n \cong R^m$, then $n = m$.

Proof. If $I \triangleleft R$ and M is an R -module, then we can consider IM which is generated by $\{rm \mid r \in I, m \in M\}$. This is an R -submodule. Look at the quotient M/IM . If $r \in I$, then

$$\begin{aligned} r \cdot - : M/IM &\rightarrow M/IM \\ m + IM &\mapsto rm + IM \end{aligned}$$

is the zero map. Therefore, we can make M/IM into an R/I -module by $(r + I) \cdot (m + IM) = rm + IM$. Let $I \triangleleft R$ be maximal (this always exists for any nonzero ring by Zorn's lemma), then both R^n/IR^n and R^m/IR^m are vector spaces over R/I . They are respectively just isomorphic to $(R/I)^n$ and $(R/I)^m$. So, $n = m$ by linear algebra since they are simply vector spaces. $\square$

19 Matrices over Euclidean domains

We aim to prove and generalise the classification of finitely generated abelian groups. A matrix A over $\mathbb{Z}$ (or any Euclidean domain) gives a homomorphism $\mathbb{Z}^m \rightarrow \mathbb{Z}^n$ (or respectively $R^m \rightarrow R^n$). The groups and modules we are interested in will be of the form $\mathbb{Z}^n / \text{im}(A)$.

Fix an Euclidean domain R and its associated Euclidean function $\phi : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$. By Euclid's algorithm, for any $a, b \in R$, we can find $x, y \in R$ such that $ax + by = \gcd(a, b)$.

Definition 19.1 – Elementary row/column operations

Let A be a matrix with entries in R . The elementary row operations are the following.

- (i) Add $c \in R$ times the i -th row to the j -th row.
- (ii) Swap the i -th row and the j -th row.
- (iii) Multiply the i -th row by a unit $c \in R^\times$.

The elementary column operations are similar.

Definition 19.2 – Equivalent matrices

Two matrices A and B are equivalent if A can be obtained from B via elementary row and column operations. Equivalently, $B = QAT^{-1}$ where Q and T are invertible.

Theorem 19.1 – Smith normal form

Any $m \times n$ matrix is equivalent to one of the form $\text{diag}(d_1, \dots, d_r, 0, \dots, 0)$ where $d_i \mid d_{i+1}$. The d_i 's are called the invariant factors of A .

Proof.

Proof. Throughout the process, we will keep calling our matrix A , even though it keeps changing in each step, so that we don't have to invent hundreds of names for these matrices.

If $A = 0$, then done! So suppose $A \neq 0$. So some entry is not zero, say, $A_{ij} \neq 0$. Swapping the i th and first row, then j th and first column, we arrange that $A_{11} \neq 0$. We now try to reduce A_{11} as much as possible. We have the following two possible moves:

- (i) If there is an A_{1j} not divisible by A_{11} , then we can use the Euclidean algorithm to write

$$A_{1j} = qA_{11} + r.$$

By assumption, $r \neq 0$. So $\phi(r) < \phi(A_{11})$ (where ϕ is the Euclidean function).

So we subtract q copies of the first column from the j th column. Then in position $(1, j)$, we now have r . We swap the first and j th column such that r is in position $(1, 1)$, and we have strictly reduced the value of ϕ at the first entry.

- (ii) If there is an A_{i1} not divisible by A_{11} , we do the same thing, and this again reduces $\phi(A_{11})$.

We keep performing these until no move is possible. Since the value of $\phi(A_{11})$ strictly decreases every move, we stop after finitely many applications. Then we know that we must have A_{11} dividing all A_{ij} and A_{i1} . Now we can just subtract appropriate multiples of the first column from others so that $A_{1j} = 0$ for $j \neq 1$. We do the same thing with rows so that the first row is cleared. Then we have a matrix of the form

$$A = \begin{pmatrix} d & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & C & \\ 0 & & & \end{pmatrix}.$$

We would like to say “do the same thing with C ”, but then this would get us a regular diagonal matrix, not necessarily in Smith normal form. So we need some preparation.

- (iii) Suppose there is an entry of C not divisible by d , say A_{ij} with $i, j > 1$.

$$A = \begin{pmatrix} d & 0 & \cdots & 0 & \cdots & 0 \\ 0 & & & & & \\ \vdots & & & & & \\ 0 & & & A_{ij} & & \\ \vdots & & & & & \\ 0 & & & & & \end{pmatrix}$$

We suppose

$$A_{ij} = qd + r,$$

with $r \neq 0$ and $\phi(r) < \phi(d)$. We add column 1 to column j , and subtract q times row 1 from row i . Now we get r in the (i, j) th entry, and we want

to send it back to the $(1, 1)$ position. We swap row i with row 1, swap column j with column 1, so that r is in the $(1, 1)$ th entry, and $\phi(r) < \phi(d)$.

Now we have messed up the first row and column. So we go back and do (i) and (ii) again until the first row and columns are cleared. Then we get

$$A = \begin{pmatrix} d' & 0 & \cdots & 0 \\ 0 & & & \\ 0 & & C' & \\ 0 & & & \end{pmatrix},$$

where

$$\phi(d') \leq \phi(r) < \phi(d).$$

As this strictly decreases the value of $\phi(A_{11})$, we can only repeat this finitely many times. When we stop, we will end up with a matrix

$$A = \begin{pmatrix} d & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & C & \\ 0 & & & \end{pmatrix},$$

and d divides *every* entry of C . Now we apply the entire process to C . When we do this process, notice all allowed operations don't change the fact that d divides every entry of C .

So applying this recursively, we obtain a diagonal matrix with the claimed divisibility property. $\square$

$\square$

Definition 19.3 – Minor

A $k \times k$ minor of a matrix A is the determinant of a $k \times k$ submatrix of A . That is, a matrix formed by removing all but k rows and k columns.

Definition 19.4 – Fitting ideal

The k -th Fitting ideal of a matrix A is $\text{Fit}_k(A) \triangleleft R$ is generated by the $k \times k$ minors of A .

Proposition 19.1

If A and B are equivalent matrices, then $\text{Fit}_k(A) = \text{Fit}_k(B)$ for all k .

Proof.

Proof. It suffices to show that changing A by a row or column operation does not change the Fitting ideal. Since taking the transpose does not change the determinant, i.e. $\text{Fit}_k(A) = \text{Fit}_k(A^T)$, it suffices to consider the row operations.

The most difficult one is taking linear combinations. Let B be the result of adding c times the i th row to the j th row, and fix C a $k \times k$ minor of A . Suppose the resultant matrix is C' . We then want to show that $\det C' \in \text{Fit}_k(A)$.

If the j th row is outside of C , then the minor $\det C$ is unchanged. If both the i th and j th rows are in C , then the submatrix C changes by a row operation, which does not affect the determinant. These are the boring cases.

Suppose the j th row is in C and the i th row is not. Suppose the i th row is $f_1, \dots, f_k$. Then C is changed to C' , with the j th row being

$$(C_{j1} + cf_1, C_{j2} + cf_2, \dots, C_{jk} + cf_k).$$

We compute $\det C'$ by expanding along this row. Then we get

$$\det C' = \det C + c \det D,$$

where D is the matrix obtained by replacing the j th row of C with $(f_1, \dots, f_k)$.

The point is that $\det C$ is definitely a minor of A , and $\det D$ is still a minor of A , just another one. Since ideals are closed under addition and multiplications, we know

$$\det(C') \in \text{Fit}_k(A).$$

The other operations are much simpler. They just follow by standard properties of the effect of swapping rows or multiplying rows on determinants. So after any row operation, the resultant submatrix C' satisfies

$$\det(C') \in \text{Fit}_k(A).$$

Since this is true for all minors, we must have

$$\text{Fit}_k(B) \subseteq \text{Fit}_k(A).$$

But row operations are invertible. So we must have

$$\text{Fit}_k(A) \subseteq \text{Fit}_k(B)$$

as well. So they must be equal. So done. □

□

Corollary 19.1

If A has Smith normal form $\text{diag}(d_1, \dots, d_r, 0, \dots, 0)$, then $\text{Fit}_k(A) = (d_1 \dots d_k)$. In particular, the d_i 's are unique up to associates.

Proposition 19.2

Let R be a principal ideal domain. Any submodule of R^m can be generated by m or fewer elements.

Proof. We proceed by induction on m .

Proof. Let $N \leq R^m$ be a submodule. Consider the ideal

$$I = \{r \in R : (r, r_2, \dots, r_m) \in N \text{ for some } r_2, \dots, r_m \in R\}.$$

It is clear this is an ideal. Since R is a principal ideal domain, we must have $I = (a)$ for some $a \in R$. We now choose an

$$n = (a, a_2, \dots, a_m) \in N.$$

Then for any vector $(r_1, r_2, \dots, r_m) \in N$, we know that $r_1 \in I$. So $a \mid r_1$. So we can write

$$r_1 = ra.$$

Then we can form

$$(r_1, r_2, \dots, r_m) - r(a, a_2, \dots, a_m) = (0, r_2 - ra_2, \dots, r_m - ra_m) \in N.$$

This lies in $N' = N \cap (\{0\} \times R^{m-1}) \leq R^{m-1}$. Thus everything in N can be written as a multiple of n plus something in N' . But by induction, since $N' \leq R^{m-1}$, we know N' is generated by at most $m-1$ elements. So there are $n_2, \dots, n_m \in N'$ generating N' . So $n, n_2, \dots, n_m$ generate N . $\square$

$\square$

Theorem 19.2

Let R be an Euclidean domain and $N \leq R^m$. There exists a basis $v_1, \dots, v_m$ of R^m such that N is generated by $d_1 v_1 + \dots + d_r v_r$ for some $r \in \{0, 1, \dots, m\}$, with $d_i \in R$ such that $d_i \mid d_{i+1}$.

Proof. By Proposition 19.2, N can be generated by $x_1, \dots, x_n$ with $n \leq m$. Consider the matrix $(x_1 \mid \dots \mid x_n)$ and put it into Smith normal form. Observe that row operations change basis for R^m and column operations change generators for N . In this new basis, N is generated by $\{d_i v_i\}_{i=1}^m$ as required. $\square$

Theorem 19.3

Let R be an Euclidean domain, any submodule of R^m is free.

Proof. In the proof of Theorem 19.3, the generating set $\{d_i v_i\}_{i=1}^m$ is linearly independent because any linear combination of the $d_i v_i$'s that gives 0 also gives one for the v_i 's themselves. So, $\{d_i v_i\}_{i=1}^m$ generate the submodule freely. $\square$

Theorem 19.4

Let R be an Euclidean domain. Let M be a finitely generated R -module. Then,

$$M \cong R \oplus \dots \oplus R \oplus \frac{R}{(d_1)} \oplus \dots \oplus \frac{R}{(d_r)}$$

where $d_i \neq 0$ and $d_i \mid d_{i+1}$.

Proof. By Lemma 17.1, there is a surjective R -module homomorphism $\gamma : R^m \rightarrow M$. We know that $\ker(\gamma) \leq R^m$. By Theorem 19.3, pick a basis $v_1, \dots, v_m$ such that $\ker(\gamma)$ is generated by $d_1 v_1 + \dots + d_r v_r$ with $d_i \mid d_{i+1}$.

By the first isomorphism theorem, $M \cong R^m / ((d_1, 0, \dots, 0), (0, d_2, 0, \dots, 0), \dots, (0, \dots, 0, d_r, 0, \dots, 0))$. This gives the result. $\square$

When $R = \mathbb{Z}$, this gives a classification of finitely generated abelian groups.

Example 19.1

Let G be an abelian group generated by a, b, c such that $2a + 3b + c = 0$, $a + 2b = 0$ and $5a + 6b + 7c = 0$. We have a surjective homomorphism $\gamma : \mathbb{Z}^3 \rightarrow G$ sending $e_1 \mapsto a$, $e_2 \mapsto b$ and $e_3 \mapsto c$. The kernel of γ is given by the image of the matrix

$$A = \begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 6 \\ 1 & 0 & 7 \end{pmatrix}.$$

By construction, $\mathbb{Z}^3 / \text{im}(A) \cong G$. To calculate the Smith normal form, we calculate the Fitting ideals of A . $\text{Fit}_1(A) = \text{Fit}_2(A) = (1)$ and $\text{Fit}_3(A) = (3)$. Hence, $G \cong \mathbb{Z}/(3)$ because $d_1 = d_2 = 1$ and $d_3 = 3$.

Proposition 19.3

Let R be an Euclidean domain with $a, b \in R$ such that $\gcd(a, b) = 1$. Then, $R/(ab) \cong R/(a) \oplus R/(b)$.

Proof. Consider

$$\begin{aligned} \varphi : R/(a) \oplus R/(b) &\rightarrow R/(ab) \\ (r_1 + (a), r_2 + (b)) &\mapsto br_1 + ar_2 + (ab). \end{aligned}$$

We can check that φ is well-defined. It is also a homomorphism. For surjectivity, we know $ax + by = 1$ for some $x, y \in R$ by the Euclidean algorithm. Observe that $\varphi((y + (a), x + (b))) = 1 + (ab)$. So, $1 + (ab)$ lies in the image of φ , which generates $R/(ab)$. For injectivity, we calculate the kernel. If $br_1 + ar_2 \in (ab)$, write $br_1 + ar_2 = abx$. Now $a \mid ar_2$ and $a \mid abx$, so $a \mid br_1$. Since $\gcd(a, b) = 1$, we have $a \mid r_1$. Similarly, $b \mid r_2$. Thus, $r_1 \in (a)$ and $r_2 \in (b)$, and the kernel is trivial. Hence, φ is an isomorphism. $\square$

Theorem 19.5 – Primary decomposition of classification

Let R be an Euclidean domain and M be a finitely generated R -module. Then, $M \cong N_1 \oplus \dots \oplus N_t$ where N_i is either R or $R/(p^n)$ for some prime p and $n \geq 1$.

Proof. We know that $M \cong R \oplus \dots \oplus R \oplus R/(d_1) \oplus \dots \oplus R/(d_r)$. Write each d_i as a product of powers of primes in R . Applying Proposition 19.3 gives the result. $\square$

20 Modules over $\mathbb{F}[X]$ and normal forms of matrices

Let V be a vector space over a field $\mathbb{F}$. Fix an endomorphism $\alpha \in \text{End}(V)$. We can make a $\mathbb{F}[X]$ -module by

$$\begin{aligned} \mathbb{F}[X] \times V &\rightarrow V \\ (f, v) &\mapsto (f(\alpha))(v). \end{aligned}$$

We will call this module V_α . If $\dim(V)$ is finite, then V_α is finitely generated as an $\mathbb{F}[X]$ -module. The vectors that finitely generate V as an $\mathbb{F}$ -module also finitely generate V_α as an $\mathbb{F}[X]$ -module.

Example 20.1

Suppose $V_\alpha \cong \mathbb{F}[X]/(X^r)$. As an $\mathbb{F}$ -vector space, $\mathbb{F}[X]/(X^r)$ has a basis $1, X, X^2, \dots, X^{r-1}$. The endomorphism $\alpha : V \rightarrow V$ acts as a multiplication by X , so

$$\begin{aligned} \mathbb{F}[X]/(X^r) &\rightarrow \mathbb{F}[X]/(X^r) \\ f \bmod X^r &\mapsto Xf \bmod X^r. \end{aligned}$$

We can write out the endomorphism in the basis $\{1, X, X^2, \dots, X^{r-1}\}$ as a matrix with ones along the subdiagonal.

Example 20.2

Similarly, if $V_\alpha \cong \mathbb{F}[X]/((X - \lambda)^r)$, then we can write $\beta : V \rightarrow V$ with $\beta = \alpha - \lambda \text{id}$. There is a basis for V where β as a matrix has ones along the subdiagonal. In this basis, α has λ 's along the diagonal and ones along the subdiagonal, which is a Jordan block.

Example. Suppose $V_\alpha \cong \mathbb{F}[X]/(f)$ for some polynomial f , for

$$f = a_0 + a_1X + \dots + a_{r-1}X^{r-1} + X^r.$$

This has a basis $1, X, X^2, \dots, X^{r-1}$ as well, in which α is represented by the matrix

$$c(f) = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{r-1} \end{pmatrix}.$$

We call this the *companion matrix* for the monic polynomial f .

These are different things that can possibly happen. Since we have already classified all finitely-generated $\mathbb{F}[X]$ modules, this allows us to put matrices in a rather nice form.

Theorem (Rational canonical form). Let $\alpha : V \rightarrow V$ be a linear endomorphism of a finite-dimensional vector space over $\mathbb{F}$, and V_α be the associated $\mathbb{F}[X]$ -module. Then

$$V_\alpha \cong \frac{\mathbb{F}[X]}{(f_1)} \oplus \frac{\mathbb{F}[X]}{(f_2)} \oplus \dots \oplus \frac{\mathbb{F}[X]}{(f_s)},$$

with $f_1 \mid f_2 \mid \dots \mid f_s$. Thus there is a basis for V in which the matrix for α is the block diagonal

$$\begin{pmatrix} c(f_1) & 0 & \dots & 0 \\ 0 & c(f_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & c(f_s) \end{pmatrix}$$

Proof. We know that V_α is a finitely-generated $\mathbb{F}[X]$ -module. By the structure theorem for these, we know

$$V_\alpha \cong \frac{\mathbb{F}[X]}{(f_1)} \oplus \frac{\mathbb{F}[X]}{(f_2)} \oplus \dots \oplus \frac{\mathbb{F}[X]}{(f_s)} \oplus 0.$$

(There can be no copies of $\mathbb{F}[X]$, since $V_\alpha = V$ is finite-dimensional over $\mathbb{F}$ but $\mathbb{F}[X]$ is not.) The divisibility criterion also follows from the structure theorem. Then the form of the matrix is immediate. $\square$

This is really a canonical form. The Jordan normal form is not canonical, since we can move the blocks around. The structure theorem determines the factors f_i up to units, and once we require it is monic, there is no choice left.

In terms of matrices, this says that if α is represented by a matrix $A \in M_{n,n}(F)$ in some basis, then A is conjugate to a matrix of the form above.

From the rational canonical form, we can immediately read off the minimal polynomial as f_s . This is since if we view V_α as the decomposition above, we find that $f_s(\alpha)$ kills everything in $\frac{\mathbb{F}[X]}{(f_s)}$. It also kills the other factors since $f_i \mid f_s$ for all i . So $f_s(\alpha) = 0$. We also know that no smaller polynomial kills V , since it does not kill $\frac{\mathbb{F}[X]}{(f_s)}$.

Similarly, we find that the characteristic polynomial of α is $f_1 f_2 \cdots f_s$.

Recall we had a different way of decomposing a module over a Euclidean domain, namely the primary decomposition: this will give us the Jordan normal form. Before we can use that, we need to know what the primes are. This is why we need to work over $\mathbb{C}$.

Lemma. The prime elements of $\mathbb{C}[X]$ are the $X - \lambda$ for $\lambda \in \mathbb{C}$ (up to multiplication by units).

Proof. Let $f \in \mathbb{C}[X]$. If f is constant, then it is either a unit or 0. Otherwise, by the fundamental theorem of algebra, it has a root λ . So it is divisible by $X - \lambda$. So if f is irreducible, it must have degree 1. And clearly everything of degree 1 is prime. $\square$

Applying the primary decomposition theorem to $\mathbb{C}[X]$ -modules gives us the Jordan normal form.

Theorem (Jordan normal form). Let $\alpha : V \rightarrow V$ be an endomorphism of a vector space V over $\mathbb{C}$, and V_α be the associated $\mathbb{C}[X]$ module. Then

$$V_\alpha \cong \frac{\mathbb{C}[X]}{(X - \lambda_1)^{a_1}} \oplus \frac{\mathbb{C}[X]}{(X - \lambda_2)^{a_2}} \oplus \cdots \oplus \frac{\mathbb{C}[X]}{(X - \lambda_t)^{a_t}},$$

where $\lambda_i \in \mathbb{C}$ do *not* have to be distinct. Thus there is a basis of V in which α has matrix

$$\begin{pmatrix} J_{a_1}(\lambda_1) & & & 0 \\ & J_{a_2}(\lambda_2) & & \\ & & \ddots & \\ 0 & & & J_{a_t}(\lambda_t) \end{pmatrix},$$

where

$$J_m(\lambda) = \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ 1 & \lambda & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & \lambda \end{pmatrix}$$

is an $m \times m$ matrix.

Proof. Apply the prime decomposition theorem to V_α . Then all primes are of the form $X - \lambda$. We then use our second example at the beginning of the chapter to get the form of the matrix. $\square$

The blocks $J_m(\lambda)$ are called the Jordan λ -blocks. It turns out that the Jordan blocks are unique up to reordering, but it does not immediately follow from what we have said so far, and we will not prove it. It is done in the IB Linear Algebra course. We can also read off the minimal and characteristic polynomials of α . The minimal polynomial is $\prod_\lambda (X - \lambda)^{a_\lambda}$ where a_λ is the size of the largest λ -block. The characteristic polynomial of α is $\prod_\lambda (X - \lambda)^{b_\lambda}$ where b_λ is the sum of the sizes of the λ -blocks. Alternatively, it is $\prod_{i=1}^t (X - \lambda_i)^{a_i}$. From the Jordan normal form, we can also read off the size of the λ -space of α , as the number of λ -blocks.